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Changes in the Synaptic Proteome in Tauopathy and Rescue of Tau-Induced Synapse Loss by C1q Antibodies

Highlights

- Pathological Tau accumulation alters protein composition of PSD in Tau-P301S mice
- Loss of postsynaptic GTPase-regulatory proteins leads to loss of dendritic spines
- C1q tags Tau-affected synapses, leading to microglial engulfment and synapse loss
- C1q-blocking antibody prevents microglial synapse removal and rescues synapse density

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In Brief

Unbiased proteomics reveals multiple molecular changes at hippocampal synapses that occur prior to neurodegeneration in a taupathy/Alzheimer's disease mouse model. Complement C1q labels phospho-Tau-containing synapses and drives microglia-mediated synapse loss that can be rescued by a C1q-blocking antibody.

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Changes in the Synaptic Proteome in Tauopathy and Rescue of Tau-Induced Synapse Loss by C1q Antibodies

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SUMMARY

Synapse loss and Tau pathology are hallmarks of Alzheimer's disease (AD) and other tauopathies, but how Tau pathology causes synapse loss is unclear. We used unbiased proteomic analysis of postsynaptic densities (PSDs) in Tau-P301S transgenic mice to identify Tau-dependent alterations in synapses prior to overt neurodegeneration. Multiple proteins and pathways were altered in Tau-P301S PSDs, including depletion of a set of GTPase-regulatory proteins that leads to actin cytoskeletal defects and loss of dendritic spines. Furthermore, we found striking accumulation of complement C1q in the PSDs of Tau-P301S mice and AD patients. At synapses, C1q decorated perisynaptic membranes, accumulated in correlation with phospho-Tau, and was associated with augmented microglial engulfment of synapses and decline of synapse density. A C1q-blocking antibody inhibited microglial synapse removal in cultured neurons and in Tau-P301S mice, rescuing synapse density. Thus, inhibiting complement-mediated synapse removal by microglia could be a potential therapeutic target for Tau-associated neurodegeneration.

INTRODUCTION

Synapse loss is a hallmark of Alzheimer's disease (AD) and other tauopathies and is central to the neurodegenerative process (Spires-Jones and Hyman, 2014). AD is a progressive neurodegenerative disease characterized by extracellular deposits of amyloid-beta (Aβ) and intracellular aggregates of abnormally phosphorylated Tau. While amyloid pathology appears long before AD symptoms manifest, it is Tau pathology and synapse loss that temporally correlate with cognitive impairment and precede overt neurodegeneration (Jack et al., 2010; Scheff et al., 2006; Spires-Jones and Hyman, 2014). Normally, Tau is enriched in axons, where it interacts with microtubules (MTs) and may regulate MT polymerization and dynamics, axonal outgrowth, and transport (Wang and Mandelkow, 2016). During progression of AD and other tauopathies, abnormally phosphorylated forms of Tau aggregate and accumulate in the somatodendritic compartment in neurons and can localize to synapses (Ittner and Götz, 2011; Tai et al., 2012). Although Tau has been observed in dendritic spines under physiological conditions (Frandemiche et al., 2014; Morris et al., 2015), infiltration of pathological Tau can disrupt synaptic functions and lead to synapse loss (Hoover et al., 2010; Zhao et al., 2016).

Despite mounting evidence that the presence of abnormal Tau is synaptotoxic (Hoover et al., 2010; Ittner et al., 2010; Zhao et al., 2016), it is not clear how Tau pathology leads to synaptic dysfunction and loss. One possibility is that pathogenic Tau may impair MT-based transport by interfering with kinesin motors (Dixit et al., 2008) or induce microtubule severing (Zempel et al., 2013), thereby hindering cargo delivery to the spine (Hoover et al., 2010). Another possibility is disrupted synaptic structure. The postsynaptic density (PSD) of excitatory synapses is usually located on dendritic spines, whose growth and stability depend on the actin cytoskeleton and whose dynamic modulation is essential for synaptic integrity and plasticity (Fukazawa et al., 2003; Okamoto et al., 2004). In AD, neuronal actin regulation might be disturbed by Tau binding and crosslinking of actin, leading to the formation of cytoplasmic bundles of filamentous actin (Fulga et al., 2007).

While neuron-autonomous disruption of synaptic composition and/or structure may contribute to synapse loss caused by Tau pathology, recent evidence in various neurodegenerative models points to an important role for synapse loss via microglial

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phagocytosis (Hong et al., 2016a; Lui et al., 2016; Vasek et al., 2016). Microglia-mediated synapse pruning is implicated in normal brain development (Hong et al., 2016b; Paolicelli et al., 2011) and involves the classical complement pathway, a part of the innate immune system that facilitates the clearance of microbes and damaged cells by phagocytes (Schafer et al., 2012; Stephan et al., 2012; Stevens et al., 2007). This developmental process of synapse elimination might be reactivated in the diseased brain (Hansen et al., 2018; Salter and Stevens, 2017). In particular, C1q, the initiator of the classical complement pathway, which is produced by microglia in the brain, colocalizes with synaptic markers and may contribute to synapse loss in several neurodegenerative models (Hong et al., 2016a; Lui et al., 2016; Vasek et al., 2016).

An unbiased examination of the effects of Tau pathology on the protein composition of synapses has not been performed. To precisely and broadly probe the molecular basis of Tau-mediated synaptic disruption, we conducted unbiased quantitative proteomics of PSD fractions in a widely used tauopathy model (hTau-P301S transgenic mice) at an early stage of disease progression before overt neurodegeneration. We identified significant alterations in a variety of proteins and pathways in PSDs from Tau-P301S mice. Pathway analysis combined with molecular experiments highlight the specific depletion of postsynaptic GTPase-regulating proteins and dysregulation of the actin cytoskeleton as a pathological mechanism of dendritic spine loss. Strikingly, the extracellular protein complement component C1q was markedly increased in PSDs of Tau-P301S mice (as well as in AD patients) and correlated with Tau pathology. We present evidence that Tau pathology induces engulfment of excitatory synapses by microglia in a C1q-dependent manner. Moreover, treatment with a C1q-neutralizing antibody reduces synapse removal by microglia and rescues synapse density in Tau-P301S mice.

RESULTS

Tau-P301S Mice Develop Cognitive and Synaptic Deficits and Accumulate Synaptic Phospho-Tau before Overt Neurodegeneration

To determine the molecular mechanisms by which pathological Tau affects synapses, we sought to identify changes in the PSD proteome early in the course of disease in transgenic mice that express the human Tau mutant Tau-P301S (Yoshiyama et al., 2007). Tau-P301S mice showed age-dependent increase in Tau pathology, as measured by phospho-Tau (AT8) immunoreactivity, especially in the hippocampus starting in the mossy fiber tract at 6 months and spreading throughout the hippocampus by 9 months (Figures 1A and 1B). AT8 staining was virtually undetectable in nontransgenic (nonTg) brains at all ages. The worsening of AT8 Tau pathology was accompanied by progressive increase in staining of reactive CD68+ microglia and GFAP+ astrocytes in the same brain regions (Figures 1A and 1B). At 12 months, Tau-P301S mice showed obvious neurodegeneration and clear hippocampal atrophy (Figure 1A). By MRI brain scan, the volume of hippocampus in Tau-P301S mice was similar to nonTg mice at 6 and 9 months of age but declined severely by 12 months of age (Figure S1A).

We assessed activity of Tau-P301S mice in the open-field assay, as hippocampal lesions are known to induce hyperlocomotion (Lecourtier et al., 2012). Tau-P301S mice behaved similarly to nonTg mice at 6 and 9 months of age but showed elevated total motor activity at 12 months of age (Figure S1B). In the Morris water maze test, which measures spatial learning and memory, 9-month-old Tau-P301S-mice learned more slowly than nonTg controls, traveling a greater distance to the target platform and spending less time in the target quadrant (Figure S1C). Electrophysiological analysis of hippocampal brain slices from 9-month-old Tau-P301S mice showed exaggerated short-term synaptic facilitation (increased paired-pulse ratio; PPR), and significantly reduced long-term potentiation (LTP) of excitatory postsynaptic potentials (EPSPs) and population spikes (Figure S1D). No significant alterations in PPR and LTP were observed in 4-month-old mice (Figure S1D), suggesting that synaptic impairment was driven by the progressive Tau pathology. Based on the above characterization of disease progression in Tau-P301S mice, we performed all subsequent analyses of PSDs at 9 months of age when robust Tau pathology and synaptic dysfunction are present, but prior to overt hippocampal neurodegeneration and atrophy.

Synaptic Enrichment of Specific Tau Species in Tau-P301S Mice

We purified PSDs from the hippocampus of nonTg and Tau-P301S mice and analyzed the nature of Tau species in PSD fractions versus total lysates by immunoblotting. While total Tau was increased in Tau-P301S mice, there was no significant difference in Tau levels in lysates versus PSD fractions within either nonTg or Tau-P301S mice when the total immunoreactivity of all Tau-bands was quantified (Figure 1C). Tau immunoblots of Tau-P301S brain, however, showed a complex banding pattern with marked differences between the PSD and total lysate, suggesting that specific Tau species were enriched in synapses (Figure 1C). As expected, overall phospho-Tau levels were highly elevated in Tau-P301S mice compared with nonTg mice. Notably, AT8, AT100, and AT180 antibodies (which recognize phosphoepitopes that are abundant in pathological aggregated and filamentous Tau of human AD) showed an elevated signal in Tau-P301S mice that was markedly enriched in PSD fractions and only weak or undetectable in nonTg mice (Figure 1D).

Correlative light-electron microscopy (CLEM) (Figure S2) revealed that hTau was localized throughout neuronal cell bodies and processes in the hippocampal CA1 region (Figures S2A–S2E). hTau was readily detectable in presynaptic termini, often associated with synaptic vesicles, and in the majority (>80%) of postsynaptic compartments close to the PSD (Figure 1E). Quantitative measurements of immunogold distribution showed that hTau is most concentrated in the region 50 nm to 250 nm from the synaptic cleft (Figure 1E). Virtually no hTau immunogold signal was detected in nonTg brain (Figure S2A). When AT8 phospho-Tau antibody was used in immunoEM, the AT8 immunogold particles were often found on dense aggregates (Figures S2F and S2G), which were also detected in presynaptic termini and in dendritic spines (Figure 1F).

Quantitative Proteomics Reveals Molecular Changes in PSDs of Tau-P301S Mice

What happens to the protein composition of PSDs as phospho-hTau protein accumulates in presynaptic and postsynaptic

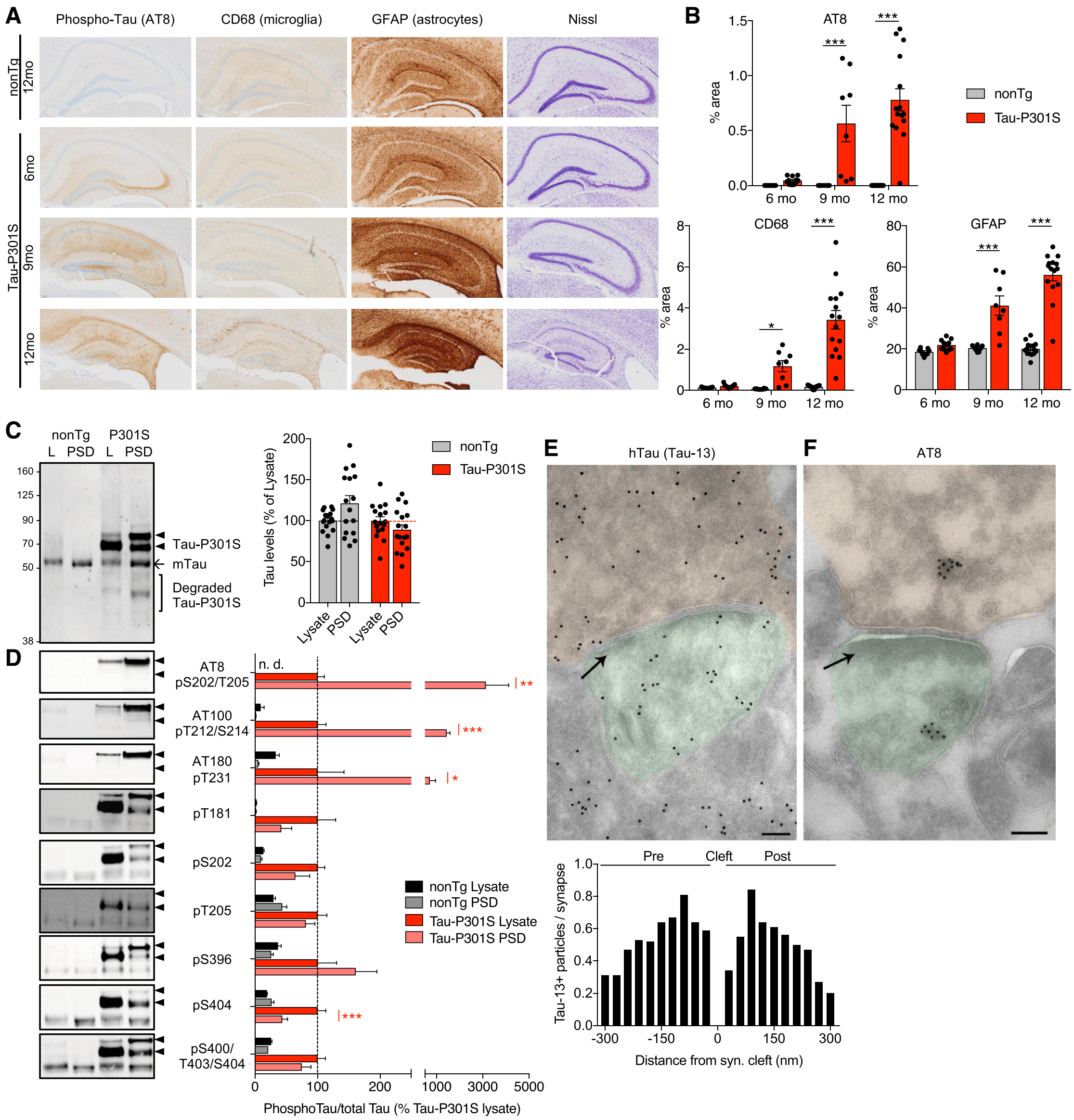

Figure 1. Progression of Tau Pathology and Accumulation of Tau at Synapses in Tau-P301S Mice

(A) Representative images of 12-month-old nonTg and 6-, 9-, or 12-month-old Tau-P301S mouse brain sections stained for phospho-Tau (AT8), activated microglia (CD68), GFAP, or Nissl.

(B) Quantification of hippocampal AT8, CD68, and GFAP immunostaining in nonTg versus Tau-P301S mice.

(C and D) Representative western blots and quantification of total Tau and phospho-Tau in hippocampal lysates and isolated PSD fractions from 9-month-old nonTg and Tau-P301S mice. Each dot in graphs of (B) and (C) represents the average value for one animal. 4–10 mice were used for phospho-Tau analysis (D).

(E) ImmunoEM of hTau in the hippocampal CA1 region. Graph shows distribution of Tau-P301S immunogold particles relative to the synaptic cleft. 20–24 synapses per mouse from 3 Tau-P301S mice were used for the EM analysis.

(F) ImmunoEM with AT8 antibodies in hippocampus of Tau-P301S mice showing aggregates of phospho-Tau. In (E) and (F), presynapses are pseudo-colored orange, postsynapses green; arrows indicate the PSD; scale bars, 100 nm.

Histograms and error bars indicate mean ± SEM. $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$; (B) two-way ANOVA with Bonferroni's multiple comparisons test; (D) one-way ANOVA with Tukey's multiple comparisons test.

See also Figure S1 and S2.

compartments? We looked for changes in the PSD fraction in 9-month-old Tau-P301S mice using label-free quantitative mass spectrometry (MS). A total of 1,257 unique proteins were identified in the PSD, of which 76 were significantly decreased and 45 significantly increased in Tau-P301S PSD fractions, relative to nonTg (Figure 2A; Table S1). Although a number of PSD proteins showed a decline in levels (Table S1), there was no significant change in the abundance of PSD scaffold molecules such as

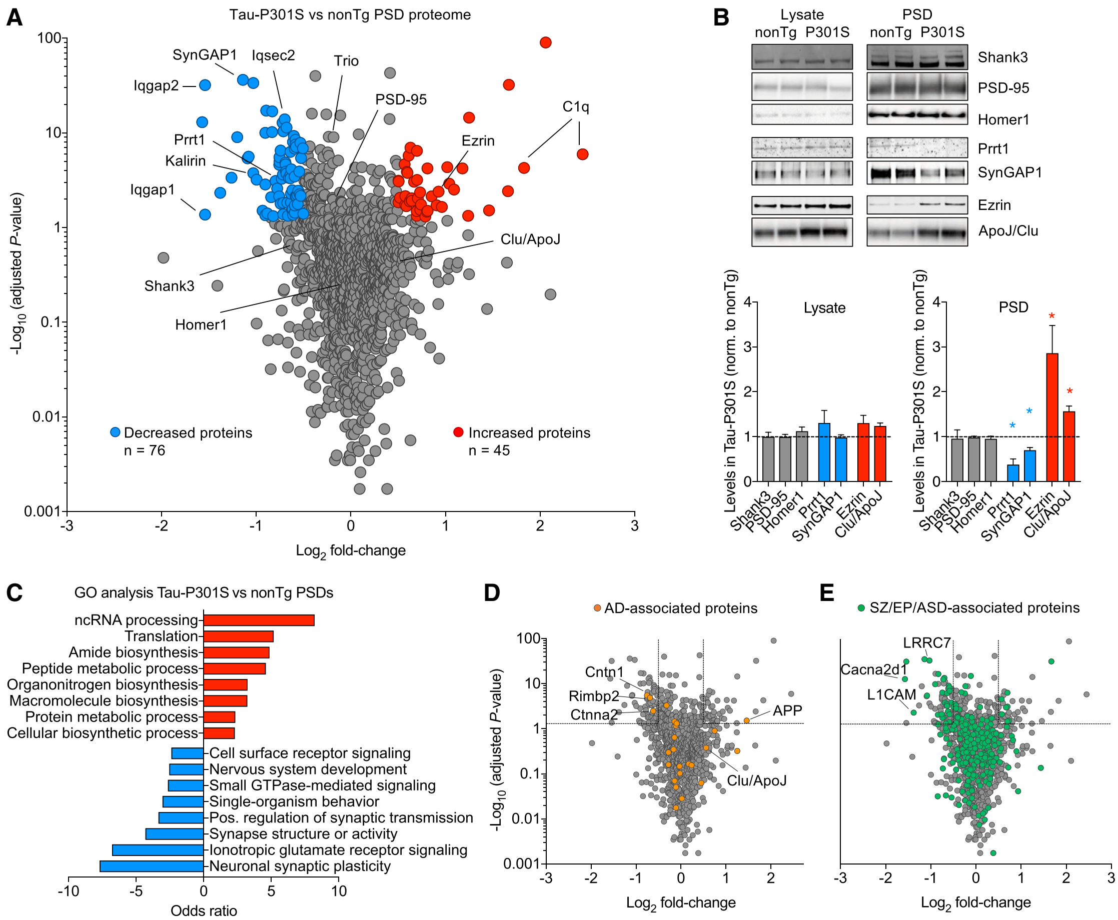

Figure 2. Unbiased Quantitative Proteomics Reveals Molecular Changes in PSDs of Tau-P301S Mice

(A) Volcano plot showing changes in proteins of isolated hippocampal PSD fractions from 9-month-old Tau-P301S versus nonTg mice, as measured by MS. The x axis specifies the $\log_2$ fold change (FC), and the y axis represents the negative $\log_{10}$ of the p values. Red and blue dots represent proteins whose abundance is significantly increased or decreased in Tau-P301S versus nonTg PSD fractions (filtering criteria: $\log_2$ FC > 0.5 and p value < 0.05). Selected labeled proteins were validated in a separate cohort in (B).

(B) Representative immunoblots and quantification of indicated proteins in hippocampal lysate and PSD fraction from nonTg and Tau-P301S mice. 3–8 mice were used for analysis.

(C) Gene ontology (GO) enrichment analysis of significantly enriched or depleted proteins in Tau-P301S versus nonTg PSD fractions. Odds ratios indicate how over- or underrepresented the differentially abundant proteins are within a particular GO category, given all of the PSD proteins detected by the MS experiment.

(D and E) Volcano plot as in (A), horizontal and vertical lines indicate the filtering criteria. Orange dots represent AD-associated proteins, and green dots represent schizophrenia/epilepsy/autism spectrum disorder (SZ/EP/ASD)-associated proteins.

Histograms and error bars indicate mean ± SEM. $^*p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$; Kruskal-Wallis test for western blots and two-sample test for equality of proportions for disease-enrichment testing.

See also Tables S1 and S2.

PSD-95, Homer1, and Shank3, which we confirmed in an independent mouse cohort by western blot (Figure 2B), suggesting that the core PSD architecture is largely intact in Tau-P301S mice at this age.

To validate the MS results, we measured a set of selected proteins in an independent mouse cohort by western blotting and confirmed that Prrt1, an AMPA-receptor-interacting protein, and SynGAP1, a synaptic Ras-GAP, were significantly decreased in Tau-P301S PSDs (Figures 2A and 2B; Table S1). Ezrin, a cross-linker of the membrane-cytoskeleton interface, was increased in the Tau-P301S PSD fractions (Figures 2A and 2B; Table S1). The total hippocampal levels of Prrt1, SynGAP1, and Ezrin were unchanged, suggesting that these proteins are specifically depleted from (Prrt1, SynGAP1) or augmented (Ezrin), in the PSD (Figure 2B). By MS, we also identified multiple post-translational modifications of hTau in the PSD, including 22 different phospho-sites, supporting the conclusion that highly phosphorylated Tau is present in the synaptic compartment in Tau-P301S hippocampi (Table S2).

Gene ontology enrichment analyses (Figure 2C) suggest that many of the downregulated biological processes in Tau-P301S PSDs are related to synapse structure and activity and glutamate receptor signaling. Indeed, several AMPA and NMDA receptor subunits were underrepresented in the Tau-P301S

PSDs (Table S1). Besides these broad processes relating to synapse structure and function, we observed a significant reduction in a set of proteins that regulate small GTPase signaling in Tau-P301S PSDs (Figures 2A and 2C). Conversely, proteins that showed enrichment in Tau-P301S PSDs are mainly involved in translation and biosynthetic processes. These include ribonucleoproteins and proteins involved in RNA processing, which might be enriched at postsynaptic sites through direct interaction with Tau (Gunawardana et al., 2015). Particularly striking was that the most highly increased proteins in Tau-P301S PSD fractions are subunits of C1q (Figure 2A; see below).

Given that many genes encoding synaptic proteins are linked to neurological and psychiatric diseases (Bayés et al., 2011), we asked whether the differentially abundant proteins in the Tau-P301S PSD fraction are associated with AD or neurodevelopmental and psychiatric disorders. We identified 24 proteins from familial AD or AD genome-wide association studies (GWASs) in the PSD proteome. Cntn1, Rimbp2, and Ctnna2, whose biology is not well understood, were significantly decreased in Tau-P301S PSDs (Figure 2D). Interestingly, amyloid precursor protein (APP) was significantly more abundant in Tau-P301S PSD fractions (Figure 2D). In addition, we noted that Clusterin/ApoJ, which shows GWAS association with late-onset AD (Harold et al., 2009), was increased ~1.5-fold in Tau-P301S PSDs by MS analysis but did not reach statistical significance (Figure 2D; Table S1). However, we could confirm by immunoblotting in an independent mouse cohort that Clu/ApoJ in PSD fractions was ~65% increased (Figure 2B). As noted previously (Noebels, 2015; Zoghbi and Bear, 2012), we found that a substantial number of PSD proteins (208/1,257) are associated with the neurodevelopmental disorders epilepsy, autism, and schizophrenia. Notably, there was a significant enrichment of such disease-associated proteins among the decreased proteins in Tau-P301S PSDs (Figure 2E; $p = 0.0008$), indicating that there might be overlapping pathophysiologic mechanisms at the synapse between tauopathies and neurodevelopmental disorders.

As a resource for further data mining, Table S1 contains the quantitative MS data of the PSD proteome for nonTg and Tau-P301S mice and the statistics of every differentially expressed protein as well as their disease association.

Loss of GTPase Regulatory Proteins and Altered Actin Cytoskeleton Could Contribute to Spine Loss in Tau-P301S-PSDs

To test the functional significance of altered levels of PSD proteins in Tau-P301S mice, we focused on the downregulated "small GTPase-mediated signal transduction" pathway. This pathway contains a number of GTPase-regulating proteins (Iqgap1, Iqgap2, Iqsec2, Kalirin, and Trio; Figure 2A), where variants in the underlying genes are associated with epilepsy, autism, schizophrenia, or intellectual disability (De Rubeis et al., 2014; Russell et al., 2014; Sadybekov et al., 2017; Shoubridge et al., 2010; Yang et al., 2014).

Iqgap1, Iqgap2, Iqsec2, Kalirin, and Trio are localized in dendritic spines of hippocampal neurons (Figure S3A) and enriched in PSD fractions of the mouse hippocampus (Figure S3B) (Gao et al., 2011; Myers et al., 2012; Xie et al., 2007). Corroborating the MS analysis, we confirmed by immunoblotting the significant decrease of PSD levels of Iqgap2, Iqsec2, and Kalirin in 9-month-old Tau-P301S hippocampi (Figure 3A). The RhoGEF Trio, which was significantly decreased in the Tau-P301S-PSD by MS analysis (but did not reach the 0.5 log2-fold change cutoff), was also significantly reduced by ~30% in Tau-P301S-PSD fractions by immunoblot analysis of an independent cohort (Figures 2A and 3A). PSD levels of these GTPase regulatory proteins were unchanged in 6-month-old mice (Figure S3C). The levels of all tested GTPase-regulating proteins were unchanged in total hippocampal lysates, suggesting that the observed reductions are specific to the postsynaptic compartment.

To examine how depletion of these GTPase-regulatory proteins affects dendritic spines, we used small interfering RNA-(siRNA) mediated knockdown in cultured hippocampal neurons (Figure S3E). Neurons incubated with individual siRNAs against Iqgap1, Iqgap2, Iqsec2, Kalirin, or Trio showed 30%–50% decreased spine density, while siRNA against Iqsec3, which was not changed in Tau-P301S PSDs, had no effect on spine density (Figure 3B). A protein-protein interaction analysis identified that Iqgap1, Iqgap2, Iqsec2, Kalirin, and Trio act in a highly interconnected network regulating the Rho family of GTPases, which in turn controls actin dynamics (Figure S3D). Thus, we hypothesized that knockdown of the GTPase regulators caused loss of spines via dysregulation of the postsynaptic actin network and investigated whether pharmacological stabilization of filamentous actin (F-actin) would stabilize spines. Neurons that had been treated with siRNAs for 5 days *in vitro* (DIV14+5), were incubated with fresh siRNAs and the F-actin stabilizing agent phallacidin for another 2 days. Phallacidin had no effect on spines in control cultures but restored spine density in neurons treated with siRNAs targeting Iqsec2, Kalirin, Trio, or Iqgap1 and Iqgap2, or even led to spine density greater than control neurons (Figure 3C). To probe the state of postsynaptic actin polymerization *in vivo*, we stained F-actin (using phalloidin) and the postsynaptic marker Homer1 in the hippocampal CA1 region in nonTg and Tau-P301S mice. The fluorescence intensity of phalloidin signal that colocalized with Homer1 clusters (a measure of the amount of postsynaptic F-actin) was reduced ~30% in Tau-P301S versus nonTg mice (Figure 3D).

Together, these results highlight that the synaptic GTPase-regulating proteins Iqgap1, Iqgap2, Iqsec2, Kalirin, and Trio are specifically decreased in the PSD compartment of Tau-P301S mice, which can lead to dysregulation of the actin cytoskeleton and potentially contribute to the spine loss seen in AD and other tauopathies.

Accumulation of Complement C1q at Synapses in Tau-P301S Mice

Among the most highly upregulated proteins in PSD fractions of Tau-P301S hippocampi was C1q (Figure 2A; Table S1). By western blotting of a different cohort of mice, we confirmed that C1q was greatly increased (with high inter-animal variability) in PSDs of 9-month-old Tau-P301S hippocampi and barely detectable in the nonTg PSD (Figure 4A). C1q levels were also increased in hippocampal lysates from Tau-P301S mice, although to a lesser degree than in PSD fractions. These biochemical data suggest

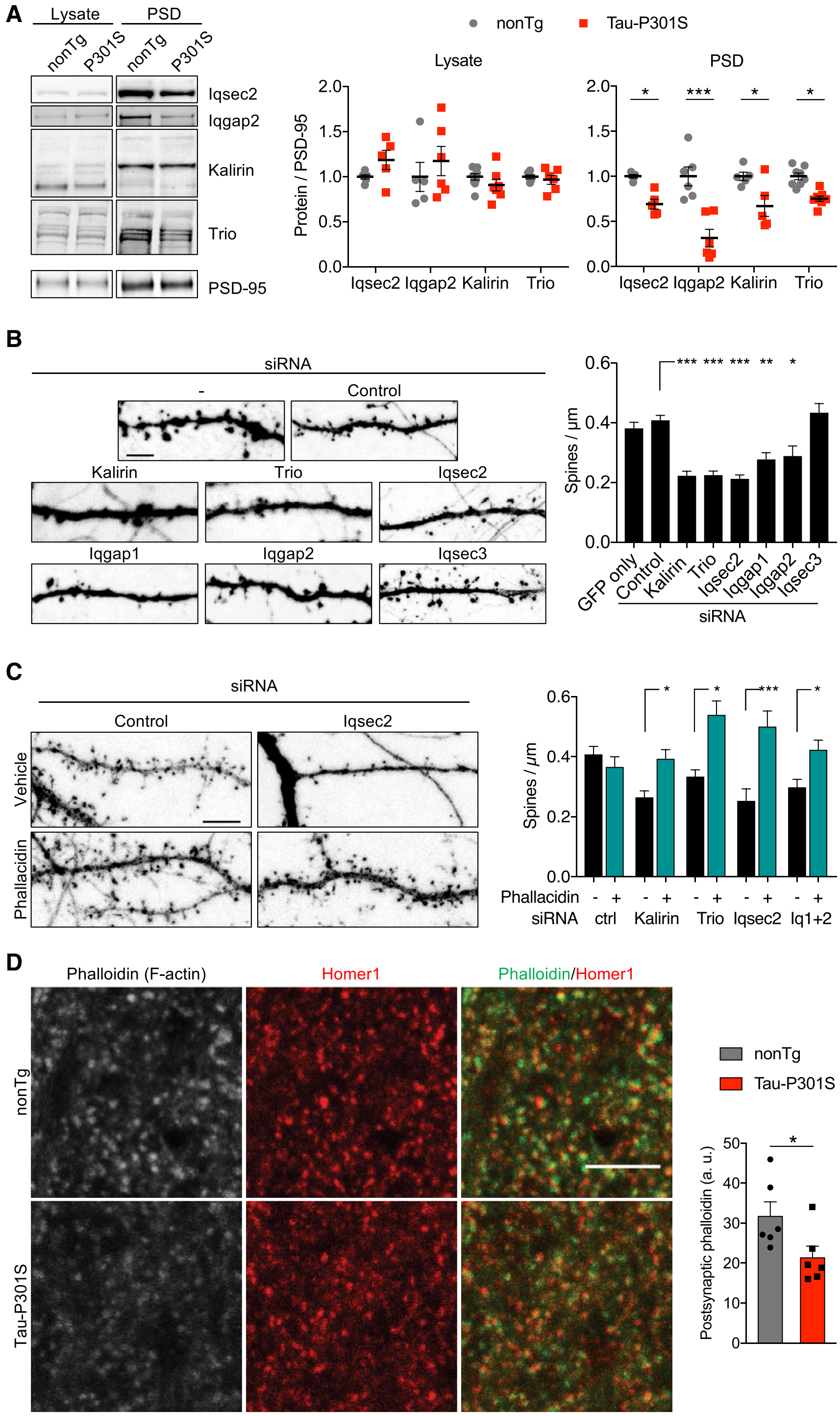

Figure 3. Small GTPase-Regulating Proteins Are Decreased in Tau-P301S-PSDs Leading to Deregulated Actin Dynamics

(A) Representative western blots and quantification of indicated proteins in hippocampal lysate and PSD fractions from 9-month-old nonTg and Tau-P301S mice. Relative expression was normalized to PSD-95 and the average of nonTg mice.

(B) Representative images of dendritic segments and quantification of spine density after treatment of GFP-expressing hippocampal neurons with indicated siRNAs for 5 days. Scale bar, 5 µm. At least 14 neurons per culture from three independent cultures were used for the analysis.

(C) Dendritic segments of GFP-expressing neurons after treatment with control or Iqsec2-specific siRNAs and incubation with phallacidin or vehicle. Scale bar, 5 µm. Histogram shows spine density along dendrites after treatment with siRNAs plus or minus phallacidin as indicated. At least 11 neurons per culture from two independent neuronal cultures were used for the analysis.

(D) Representative fluorescent image of phalloidin and Homer1 staining in hippocampal CA1 region. Quantitation of intensity of phalloidin signal that overlaps with Homer1 puncta. Scale bar, 5 µm. Each dot represents the average for one mouse.

Histograms and error bars indicate mean ± SEM. *$p < 0.05$, **$p < 0.01$, ***$p < 0.001$; (A) and (C) two-way ANOVA with Bonferroni's multiple comparisons test; (B) one-way ANOVA with Tukey's multiple comparisons test; (D) two-tailed Student's t test.

See also Figure S3.

that C1q accumulates at synapses in the Tau-P301S brain before overt neurodegeneration has taken place.

Stimulated emission depletion (STED) super-resolution microscopy of brain sections, co-stained for C1q and presynaptic (Synapsin) and postsynaptic (PSD-95) markers (Figure 4B; Figure S4A), showed that a significant proportion of C1q immunoreactive puncta is located at synapses (Figure 4B; Figure S4A). The percentage of C1q-labeled synapses in hippocampal CA1 was much higher in Tau-P301S (18.2% ± 0.9%) than in non-Tg brains (8% ± 1.6%) at 9 months of age (Figure 4C), corroborating the biochemical results from PSD isolations. Moreover, using CLEM with immunogold labeling for C1q, we observed increased density of synaptic C1q labeling in the dentate gyrus (DG) of Tau-P301S mice (Figure 4D). C1q immunogold particles were not localized in the synaptic cleft or at the synapse proper, but rather were found in perisynaptic regions associated with the pre- and postsynaptic membranes. Intriguingly, the increase in synaptic C1q labeling in Tau-P301S mice was mainly driven by augmented postsynaptic C1q, as the percentage of presynaptic labeling by C1q was only slightly increased between Tau-P301S and nonTg mice (Figure 4D).

Although C1q levels in the PSD were highly variable between individual Tau-P301S animals (Figure 4A), there was a strong positive correlation between the amount of C1q and

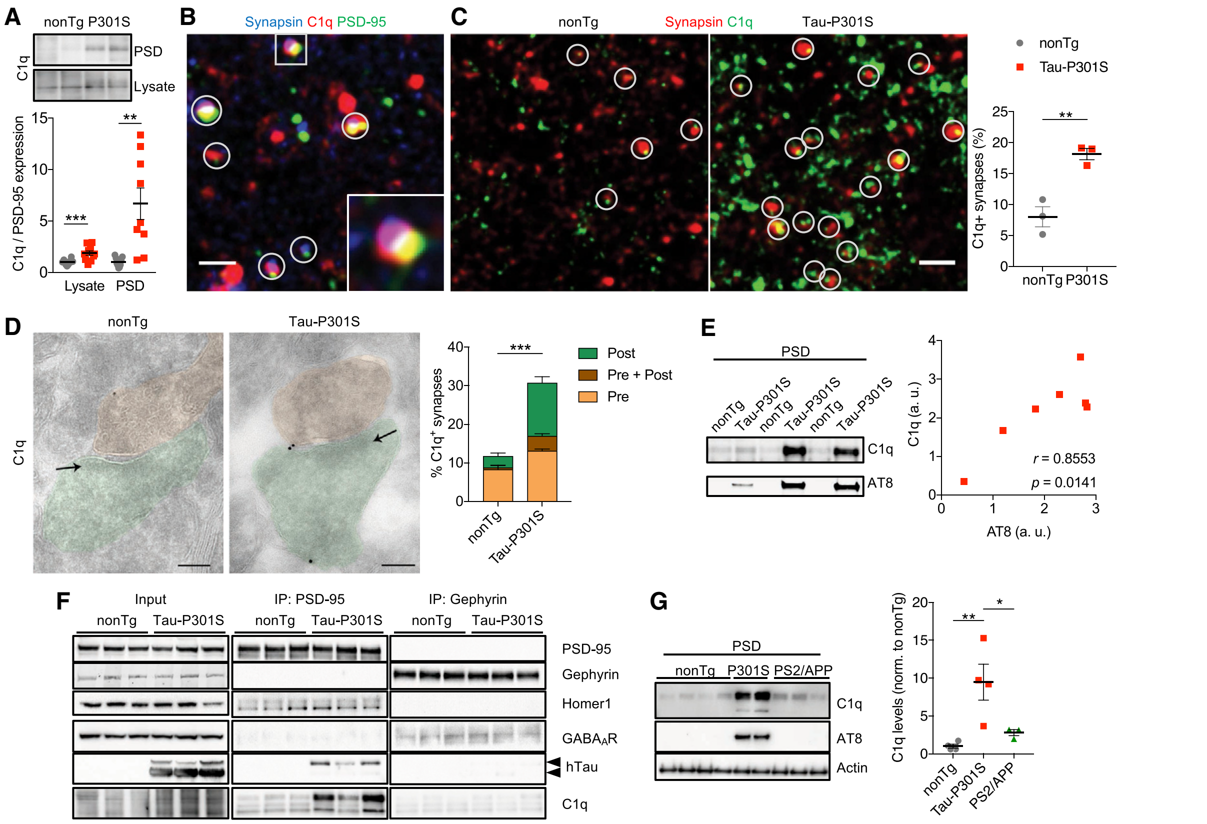

Figure 4. C1q Accumulates at Synapses in Tau-P301S Hippocampus

(A) Representative immunoblot and quantification of C1q in hippocampal PSD fractions or lysates from nonTg and Tau-P301S mice. C1q levels were normalized to PSD-95 and the average of nonTg mice.

(B) Representative STED image of Synapsin (blue), C1q (red), and PSD-95 (green) in CA1 region of 9-month-old Tau-P301S mouse. Circles and square show examples of C1q puncta colocalized with Synapsin and PSD-95 puncta. Square is enlarged in inset; see also Figure S4A for quantification of colocalization. Scale bar, 500 nm.

(C) Synapsin (red) and C1q (green) puncta in STED images from CA1 region of nonTg and Tau-P301S mice. Co-localized Synapsin and C1q puncta are indicated by circles. Scale bar, 1 μm. Graph shows percent of total Synapsin clusters that colocalized with C1q in nonTg versus Tau-P301S. At least 6,000 synapses per mouse were analyzed.

(D) Representative immunoEM images of synaptically localized C1q in dentate gyrus. Presynapses are pseudo-colored in orange, postsynapses in green. Arrows indicate the PSD. Plot shows the percentage of C1q-labeled synapses, with colored partitions indicating the percent of synapses in which C1q label was localized at presynaptic (orange), postsynaptic (green), or both pre- and postsynaptic sites (brown). Scale bars, 100 nm. 69–108 synapses per mouse from 3 animals per genotype were analyzed.

(E) C1q and AT8 immunoblots of PSD fractions from the nonTg and Tau-P301S mice, and correlation of C1q and AT8 level in PSD fractions of Tau-P301S mice (a.u., arbitrary units).

(F) Co-immunoprecipitation analysis of hTau and C1q with PSD-95 or Gephyrin antibodies from nonTg or Tau-P301S hippocampal lysates. Arrowheads indicate the positions of the two major hTau bands.

(G) Representative immunoblots and quantitation of C1q and AT8 levels in PSD fractions from 9-month-old nonTg, Tau-P301S, and PS2/APP mice hippocampi. Histograms and error bars indicate mean ± SEM. Each dot represents the average for one mouse. $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$; (A), (B), and (D) two-tailed Student's t test, (G) one-way ANOVA with Tukey's multiple comparisons test.

See also Figure S4.

the amount of AT8 phospho-Tau in PSD preparations from different Tau-P301S mice (Figure 4E), consistent with the idea that Tau pathology causes C1q accumulation and that C1q associates with synapses that have been affected by phospho-Tau. Importantly, antibodies against PSD-95 (but not Gephyrin, the scaffold of GABAergic synapses) co-immunoprecipitated C1q and hTau from hippocampal lysates, confirming that C1q and Tau are specifically in the PSD-95 complex associated with excitatory synapses (Figure 4F). In Tau-P301S hippocampi, only the higher-migrating phosphorylated Tau band co-immunoprecipitated with PSD-95 (Figure 4F), further suggesting that it is the pathological phospho-Tau that interacts with the PSD. The amount of C1q co-immunoprecipitated with PSD-95 was greater from Tau-P301S than from nonTg hippocampus and mirrored the amount of co-immunoprecipitated phospho-Tau, supporting the idea that C1q associates with phospho-Tau-containing synapses (Figure 4F).

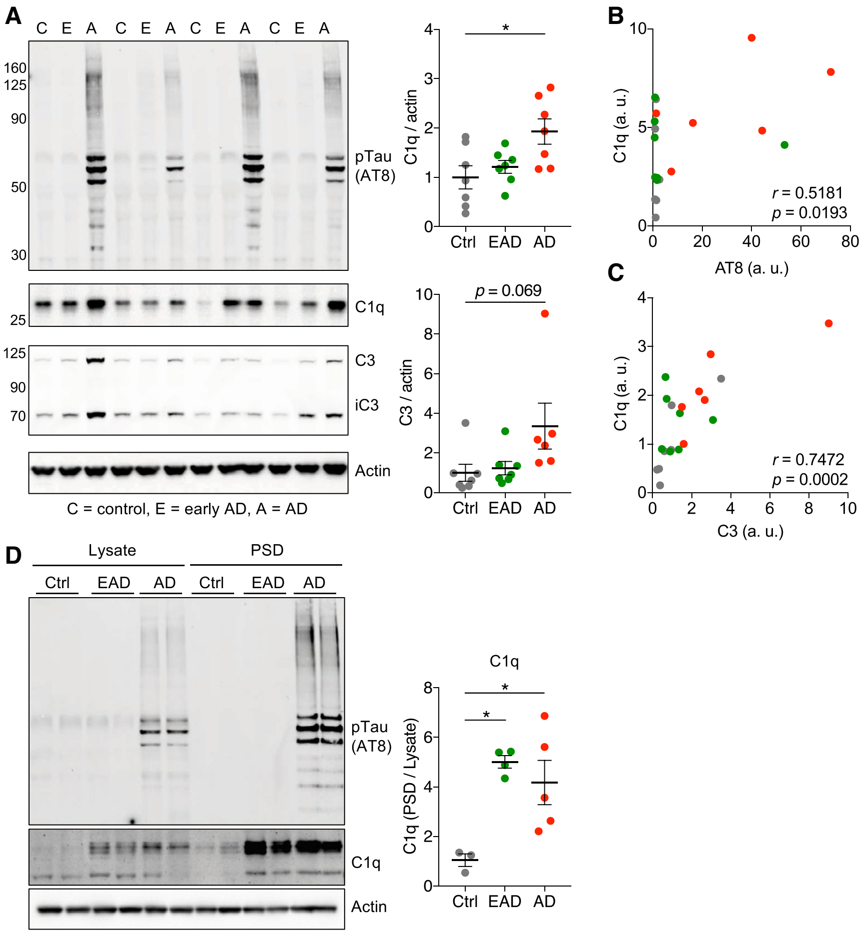

Figure 5. Increased Complement Levels in Brain Tissue and PSDs from Human AD

(A) Immunoblots for the indicated proteins in superior frontal gyrus (SFG) total lysates from control, early AD (EAD), and AD brains. Quantification of C1q and C3 levels in patient lysates, normalized to actin and the average of control.

(B) Correlation between C1q and AT8 levels in SFG lysates.

(C) Correlation between C1q and C3 levels in SFG lysates (a.u., arbitrary units).

(D) Immunoblots for C1q and AT8 in lysates and PSD fractions isolated from SFG of control, EAD, and AD human brains. Quantitation of PSD enrichment of C1q.

Histograms and error bars indicate mean ± SEM. $^*p < 0.05$; (A) and (D) one-way ANOVA with Tukey's multiple comparisons test; (B) and (C) two-tailed Student's t test.

See also Figure S5 and Table S3.

During disease progression, C1q signal was prominent in the hippocampus already at 6 months, especially in the DG and CA2/3 region, in both Tau-P301S and nonTg mice (Figure S4B). In nonTg mice, hippocampal C1q immunostaining stayed roughly constant from 6–12 months of age, whereas in Tau-P301S mice it increased in intensity at 9 and 12 months (Figure S4B). Increased hippocampal C1q immunoreactivity lagged slightly behind phospho-Tau pathology insofar as AT8 staining emerged above nonTg levels by 6 months of age and plateaued at 9–12 months (Figures 1A and 1B), whereas C1q intensity rose above nonTg levels only at 9 months and continued to climb through 12 months of age (Figure S4B).

Increased C1q levels have recently been reported in mouse models of β-amyloidosis (Hong et al., 2016a), so we compared the PSD level of C1q present in Tau-P301S mice to that of PS2/APP transgenic mice (expressing mutant Presenilin 2 [PS2] N141I and APP Swedish mutation [K595N/M596L]) at 9 months of age when there are abundant amyloid plaques in the brain (Richards et al., 2003). The average level of C1q in the PSD of Tau-P301S mice was elevated ~9-fold above nonTg controls, whereas it was increased only 2- to 3-fold in the PSD of PS2/APP mice (Figure 4G). Thus, at least in these mouse models, Tau pathology seems to more potently induce C1q accumulation at synapses than does β-amyloidosis. There was slightly more microgliosis associated with P301S-tauopathy than with PS2/APP amyloidosis (Figure S4), and given that microglia are a major source of C1q production in the brain, it is possible this difference might contribute to the greater C1q accumulation in the Tau-P301S brain.

Complement Is Increased and Associated with Synapses in Human AD Brains

Does C1q also accumulate and associate with synapses in human brain tissue from AD patients? Although variable between individuals, mean levels of C1q in superior frontal gyrus (SFG) lysates were increased ~1.5-fold in patients with a diagnosis of early-stage AD (EAD) and ~2-fold elevated in patients with autopsy-confirmed AD (Figure 5A; Table S3). Like in the Tau-P301S mice, C1q levels were significantly correlated with phospho-Tau (AT8) levels (Figure 5B). We also analyzed the downstream complement factor C3, which was unchanged in SFG lysates of EAD patients but increased ~2-fold in AD patients (Figure 5A) and significantly correlated with C1q levels among the individuals (Figure 5C).

We isolated PSDs from a small number of SFG tissues from AD patients and assessed whether phospho-Tau and C1q are biochemically associated with synapses. AT8 immunoreactivity was enriched in PSD fractions compared to total lysates from AD brains, similar to Tau-P301S mice. C1q was elevated in total lysate and highly enriched in the PSD fractions from EAD as well as AD patients (Figure 5D). These data indicate that C1q is enriched in human AD brains and appears to be associated with synapses in EAD and AD.

We measured the expression of complement genes (all genes from the complement pathway as well as complement regulatory proteins and receptors) by RNA sequencing of fusiform gyrus from normal controls and autopsy-confirmed AD patients and cross-checked it with an independent microarray dataset from

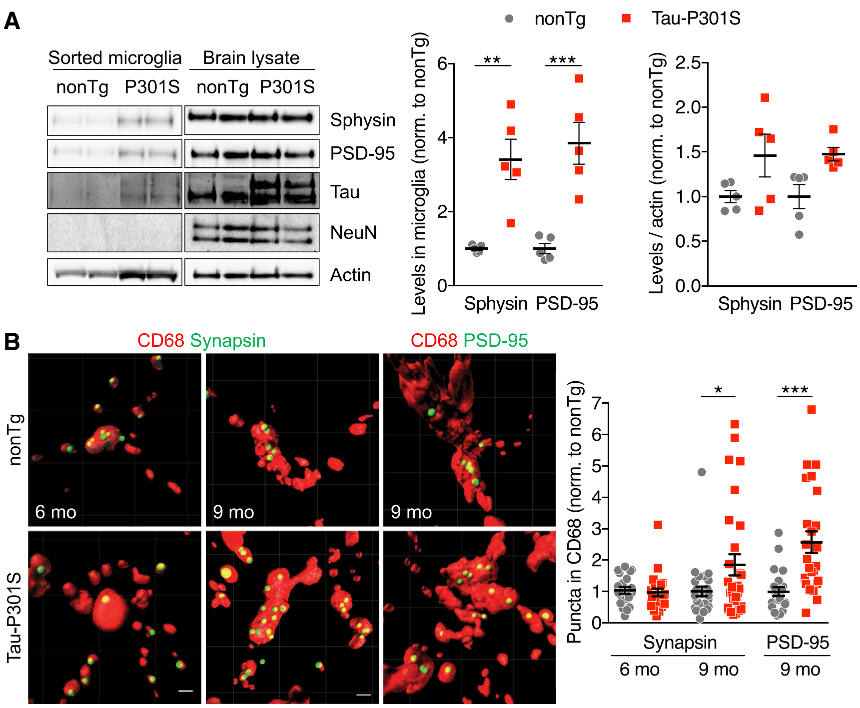

Figure 6. Microglial Engulfment of Synapses in Hippocampus of Tau-P301S Mice

(A) Representative immunoblots for the indicated proteins in Iba1+ FACS-sorted microglia from forebrain of nonTg and Tau-P301S mice. Quantitation of Synaptophysin (Sphysin) and PSD-95 immunoblots, normalized to nonTg microglia, or normalized to actin and the average of nonTg microglia. Each dot represents one mouse.

(B) 3D rendering of confocal stacks of CD68+ microglial lysosomes and engulfed PSD-95 or Synapsin puncta in CA1 region of nonTg and Tau-P301S mice at 6 or 9 months of age. Quantitation of relative number of Synapsin and PSD-95 puncta in CD68+ lysosomes normalized to nonTg mice. 4–6 animals were used for microglial engulfment analysis, each dot is data from one microglia. Scale bar, 1 μm.

Histograms and error bars indicate mean ± SEM. *$p < 0.05$, **$p < 0.01$, ***$p < 0.001$; two-way ANOVA with Bonferroni's multiple comparisons test.

See also Figure S6.

temporal cortex (Friedman et al., 2018; Webster et al., 2009) (Figure S5A). Expression of the complement gene set was significantly increased in AD patients in both data sets (Figure S5B). The transcriptomic data supports the idea that the complement pathway is elevated in AD brains.

Microglial Engulfment of Synapses Is Increased in P301S-Transgenic Mice

Labeling of synapses by C1q is thought to lead to engulfment by microglia, which might be a driver of synapse loss (Hong et al., 2016a). Indeed, we detected a 25%–30% reduction in synapse density (as defined by Synapsin or PSD-95 puncta density) in the CA1 region of 9-month-old Tau-P301S mice (Figure S6A). Synapse density was similar between Tau-P301S and nonTg animals at 6 months (Figure S6A), an age when C1q and AT8 staining show little change between nonTg and Tau-P301S hippocampus (Figure 1B; Figure S4B). Consistent with a slight loss of synapses at 9 months, the protein yield of biochemically isolated PSDs was ~20% lower from Tau-P301S hippocampi (Figure S6B).

To investigate whether microglia engulf neuronal material in disease, we immunoisolated CD11b+ microglia from the forebrain of nonTg and Tau-P301S mice by flow cytometry (Srinivasan et al., 2016) and analyzed the sorted microglia by western blotting. In these highly purified microglia, we detected low levels of Synaptophysin and PSD-95 in nonTg mice, which were increased ~3- to 4-fold in Tau-P301S mice (Figure 6A). When normalized to actin, which was consistently elevated in the microglia prep purified from Tau-P301S brains, Synaptophysin and PSD-95 levels were increased only ~1.5 fold (Figure 6A), suggesting that the increase can be explained in large part by an increased number and/or size of microglial cells in Tau-P301S mice. Tau was readily detectable in microglia isolated from Tau-P301S, but not nonTg brain. NeuN, a neuronal marker found in the nucleus, was undetectable in any of the purified microglia (Figure 6A), suggesting that microglia engulf synapses and perhaps not neuronal soma and arguing against a non-specific contamination of the microglial prep by neuronal cells.

To measure synapse engulfment by microglia more directly, we used immunofluorescence confocal microscopy to quantify PSD-95 and Synapsin puncta present within microglial lysosomes (visualized using microglial lysosome marker CD68) in brain sections (Figure 6B). In line with the biochemical data from isolated microglia, we observed an ~2-fold increase in the number of PSD-95 and Synapsin puncta that are located within microglial lysosomes in hippocampal CA1 region from Tau-P301S mice at 9 months, but not at 6 months, of age (Figure 6B). Taken together, the biochemical and imaging data show that there is increased microglial engulfment of synapses (both pre- and postsynaptic compartments) in Tau-P301S mice at 9 months of age, temporally correlating with accumulation of C1q and a drop in synapse density.

C1q-Blocking Antibody Decreases Microglial Synapse Engulfment *In Vitro*

Based on the above findings, we hypothesized that C1q-dependent microglial engulfment of synapses is a crucial mechanism leading to synapse loss in tauopathy. Would inhibition of C1q function be sufficient to prevent synapse removal by microglia? To test this, we used a C1q-blocking antibody, which potently blocked C1q binding to cultured neurons (Figures S7A and S7B). We first tested whether the anti-C1q antibody could decrease microglial synapse engulfment *in vitro.* Primary rat hippocampal neurons were transfected with Tau-P301S at DIV7; a week later (DIV14), they were co-cultured with microglia (3:1 neuron:microglia ratio) for another 3 days. Compared with dendrites of neurons cultured in absence of

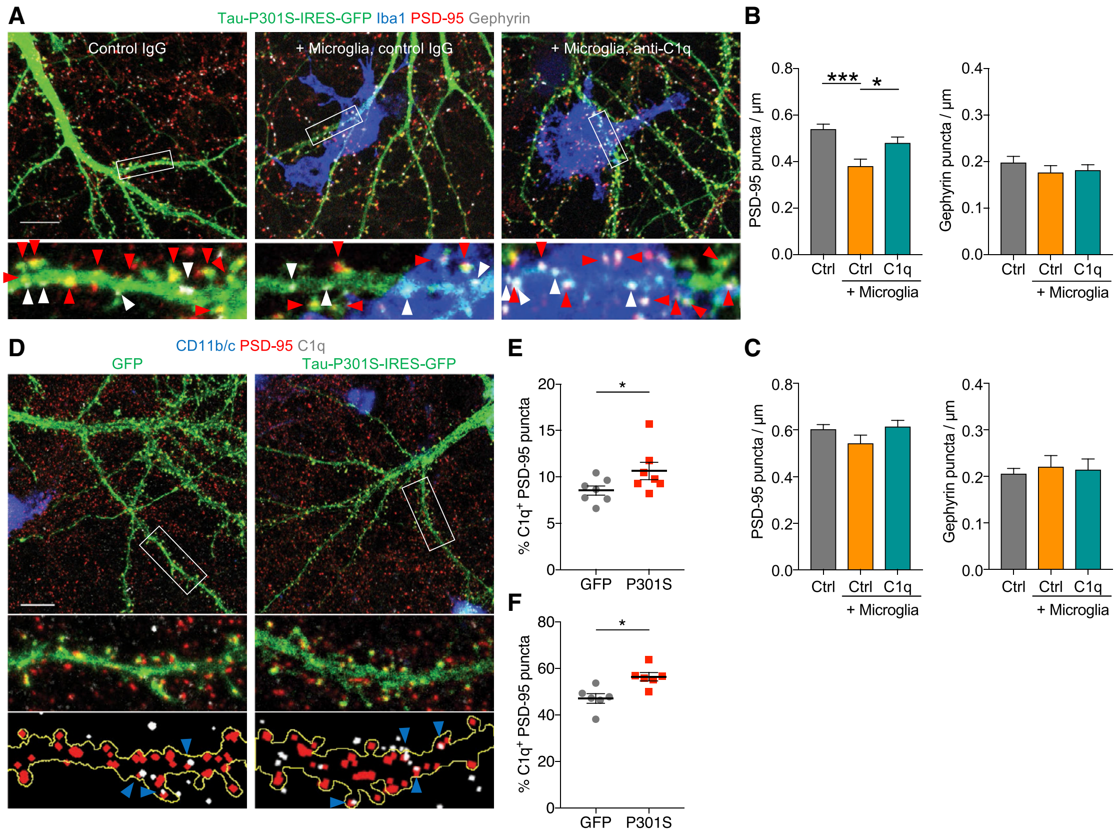

Figure 7. C1q-Blocking Antibody Inhibits Microglial Engulfment and Synapse Loss in Microglia-Neuron Co-cultures

(A) Representative images of dendritic segments in neuron-microglia co-cultures treated with isotype control IgG or anti-C1q antibodies. Neurons were transfected with a Tau-P301S-IRES-GFP construct (green), and cultures were stained for Iba1 (blue), PSD-95 (red), and Gephyrin (white). Red and white arrowheads indicate examples of PSD-95 and Gephyrin puncta, respectively. Scale bar, 10 μm.

(B and C) Quantification of dendritic PSD-95 and Gephyrin puncta density in Tau-P301S-IRES-GFP (B) or GFP-expressing (C) neuronal or neuron-microglia co-cultures after treatment with control IgG or anti-C1q antibodies. Three independent cultures were used for analysis.

(D) Representative images of neuron-microglia co-cultures stained for CD11b/c (blue), PSD-95 (red), and C1q (white). Neurons were transfected with Tau-P301S-IRES-GFP or GFP (green), as indicated. Blue arrowheads indicate colocalized C1q+ PSD-95 puncta. Scale bar, 10 μm.

(E) Percentage of PSD-95 puncta that are C1q+ in neuron-microglia co-cultures.

(F) Percentage of C1q+ PSD-95 puncta in GFP- or P301S-IRES-GFP-transfected neurons incubated with exogenous human C1q. Each dot represents the average value from 10–15 neurons from one culture.

Histograms and error bars indicate mean ± SEM. *$p < 0.05$, ***$p < 0.001$; (B) one-way ANOVA with Tukey's multiple comparisons test; (E) and (F) two-tailed Student's t test.

See also Figure S7.

microglia, synapse density, as measured by PSD-95 clusters, was reduced by ~30% on "microglia-proximal" dendrites (overlapping or in contact with Iba1$^+$ cells) but was unchanged on "microglia-distant" dendrites (>30 μm separated from Iba1$^+$ cells) (Figure S7C), suggesting that removal of synapses is mediated through close neuron-microglia interaction rather than soluble neurotoxic factors secreted by microglia. To test whether C1q is important for depletion of synapses by microglia, we added the C1q-blocking antibody or IgG isotype control at the same time as microglia to the neurons (Figure 7A). C1q-blocking antibodies, but not IgG isotype control, prevented the loss of PSD-95 puncta along microglia-proximal dendrites (Figures 7A and 7B). The density of Gephyrin clusters was not affected by microglia or by addition of C1q antibodies, further suggesting that microglia selectively remove excitatory synapses without damaging the dendrite (Figures 7A and 7B). In control neurons expressing GFP only, PSD-95 puncta density was not significantly affected by co-culture with microglia (Figure 7C), indicating that synapse pruning by microglia is dependent on Tau-P301S expression. In line with the hypothesis that synapse removal involves C1q, the percentage of C1q-labeled synapses was significantly higher on neurons transfected with Tau-P301S than on control GFP-expressing neurons in neuron-microglia cultures (Figures 7D and 7E). Addition of exogenous C1q also resulted in more C1q labeling of synapses on Tau-P301S transfected neurons (Figure 7F). Thus, overexpression of Tau-P301S in neurons appears to enhance the binding of C1q to synapses.

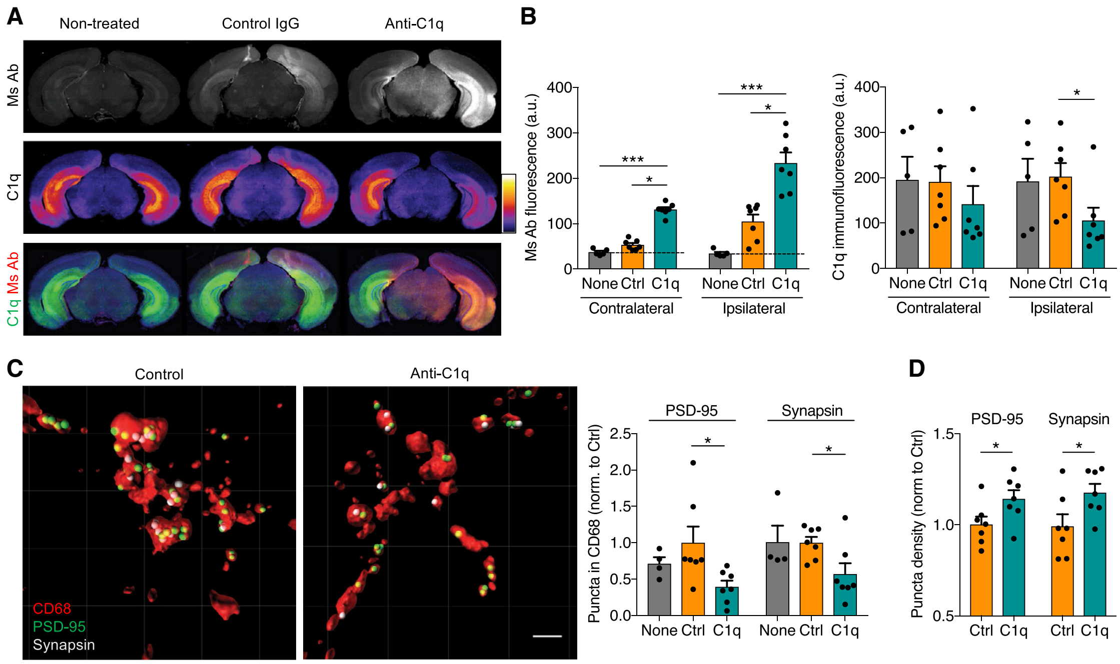

Figure 8. C1q-Blocking Antibody Inhibits Microglial Engulfment and Rescues Synapse Density in Tau-P301S Mice

(A) Immunostaining for mouse (Ms) antibodies and C1q in Tau-P301S mice injected in the right dorsal hippocampus with control IgG or anti-C1q antibodies, compared with animals that did not undergo the surgery (Non-treated).

(B) Quantitation of immunofluorescence intensity of injected antibodies and of C1q in ipsilateral and contralateral hippocampi. Each dot is the average value from one mouse.

(C) 3D rendering of confocal stacks of $CD68^+$ lysosomes (red) and engulfed PSD-95 (green) and Synapsin (white) puncta in CA1 region of Tau-P301S mice injected with control IgG or anti-C1q. Scale bar, 2 μm. Quantitation of relative number of engulfed PSD-95 and Synapsin puncta in CD68+ lysosomes normalized to control IgG-injected mice.

(D) Density of PSD-95 and Synapsin puncta in CA1 region from control IgG or anti-C1q injected Tau-P301S mice. Each dot represents the average value for one mouse.

Histograms and error bars indicate mean ± SEM. *$p < 0.05$, ***$p < 0.001$; (B) and (C) Kruskal-Wallis test; (D) two-tailed Student's t test.

See also Figure S8.

C1q-Blocking Antibody Decreases Microglial Synapse Engulfment and Rescues Synapse Density in Tau-P301S Mice

To determine whether C1q-blocking antibodies can prevent synapse engulfment by microglia and rescue synapse loss *in vivo*, we injected C1q-blocking or isotype control antibodies into the right dorsal hippocampus of 9-month-old Tau-P301S mice and analyzed the animals 5 days later (Figures 8A and 8B). The injected anti-C1q antibodies were present throughout the ipsilateral hippocampus and, to a lesser extent, the contralateral side (Figures 8A and 8B). The distribution pattern of anti-C1q antibody was similar to that of C1q protein, suggesting that the antibody bound the target (Figure 8A). Control IgG showed diffuse staining in hippocampus and cortex (ipsilateral > contralateral) with no specific colocalization with C1q (Figures 8A and 8B). In brains injected with anti-C1q, the intensity of C1q staining was reduced in the ipsilateral hippocampus, while control IgG injection had no effect (Figures 8A and 8B). In the ipsilateral CA1 region, the C1q-blocking antibody resulted in ~50% reduction in PSD-95 or Synapsin puncta within microglial lysosomes, as compared to control IgG (Figure 8C), suggesting that C1q neutralization inhibited microglial synapse engulfment in Tau-P301S brains. Remarkably, we measured a slight but significant increase in PSD-95 and Synapsin puncta density in CA1 at 5 days after injection of C1q blocking antibody, relative to control IgG (Figure 8D). Although electrophysiological properties of acute hippocampal slices (including LTP) were not changed at 5 days after injection of C1q antibodies (Figure S8), our results demonstrate that C1q antibodies can reduce synapse removal by microglia and lead to a recovery of synapse density in cultures and *in vivo*.

DISCUSSION

Synapse loss is one of the best correlates of cognitive deficits in AD, but its underlying molecular mechanism is poorly understood (Sheng et al., 2012; Spires-Jones and Hyman, 2014). Synapse loss correlates more closely with Tau pathology in the AD brain than with the load of amyloid plaque, which starts accumulating many years prior to onset of disease symptoms (Weiner et al., 2015). Moreover, there is mounting evidence that Tau pathology and neurotoxicity is, at least in part, downstream of amyloidosis in the pathogenic cascade underlying AD (Choi et al., 2014; Ittner and Götz, 2011; Musiek and Holtzman, 2015; Roberson et al., 2007). For these reasons, we focused on tauopathy and the molecular mechanisms by which tauopathy drives synapse loss. At a disease stage before obvious neurodegeneration, MS analysis showed that whereas the core architecture of the PSD, which is built around scaffolds like PSD-95 (Chen et al., 2011; Sheng and Hoogenraad, 2007), appears largely intact, multiple changes occur in the hippocampal PSD proteome. The biochemical analysis of the PSD provides only a view of the "average" synapse and so may hide the heterogeneity of changes in individual PSDs that occur in response to tauopathy. With that caveat noted, PSDs of tauopathy-affected neurons are depleted of a specific set of intracellular signaling proteins with known postsynaptic functions (small GTPase regulators) while accumulating an extracellular protein involved in innate immunity (complement C1q).

Iqgap1, Iqgap2, Iqsec2, Kalirin, and Trio regulate Rho family GTPases and hence play a pivotal role in the dynamics of the F-actin cytoskeleton in spines (Spence and Soderling, 2015). Loss of these GTPase regulators could provide a molecular explanation for why actin is compromised in spines of AD patients (Kommaddi et al., 2018). Because these proteins normally regulate dendritic spine integrity and synapse density as well as synaptic signaling, the PSD-specific depletion of Iqgap1, Iqgap2, Iqsec2, Kalirin, and Trio could contribute to synaptic dysfunction and cognitive impairment in AD, as evidenced by synaptic and/or cognitive deficits upon deletion of these genes (Cahill et al., 2009; Elagabani et al., 2016; Gao et al., 2011; Herring and Nicoll, 2016). It is intriguing that this set of GTPase regulators, as well as a disproportionate number of the other downregulated PSD proteins in tauopathy, are genetically linked to neurodevelopmental disorders such as autism spectrum disorder and schizophrenia (Fromer et al., 2014; Li et al., 2016; Noebels, 2015; Shoubridge et al., 2010; Zoghbi and Bear, 2012). Thus, we speculate that, despite different etiologies and disparate ages of onset of AD versus neurodevelopmental disorders, overlapping alterations in biochemical and cell biological signaling mechanisms may occur at the synapse to mediate the synaptic/circuit dysfunction and cognitive symptoms that are in common between neurodegenerative and neurodevelopmental diseases. Further underscoring this "shared mechanism" concept is the recent discovery that genetic variants of the complement component *C4A* locus (which are associated with elevated expression of C4) increase the risk of schizophrenia (Sekar et al., 2016), and that complement may play a role in synapse elimination in neurodevelopmental disorders (Neniskyte and Gross, 2017; Zoghbi and Bear, 2012), as well as in neurodegeneration.

Our unbiased identification of C1q accumulation in Tau-P301S PSDs, the close correlation between synaptic levels of C1q and phospho-Tau, and the high-resolution localization of C1q at excitatory synapses by light microscopy (LM) and electron microscopy (EM) provide compelling evidence that tauopathy induces the C1q tagging of synapses. Given that C1q is a secreted protein produced mainly by microglia, and microglia engulf synapses more avidly in the Tau-P301S brain, our study points to a critical microglia-neuron interaction in the synapse loss of tauopathy. We also found elevated C1q in AD patient brains, not only in bulk tissue, but also in purified PSD fractions, consistent with synaptic tagging by C1q in AD. Our findings support the idea that the complement pathway, which plays a role in synaptic pruning during development in various regions of the brain, is re-activated in neurodegenerative disease (Stephan et al., 2012).

Some interesting points are revealed by this study. (1) Tauopathy seems more potent than amyloidosis in inducing C1q accumulation at synapses; this difference correlates with the strong neurodegeneration phenotype of Tau-P301S mice (see Figure 1) versus the minimal neurodegeneration phenotype of PS2/APP transgenic mice (Richards et al., 2003) and supports the idea that Tau-mediated complement activation is a key mechanism in the synapse loss and neurodegeneration of AD. However, while tauopathy seems to be sufficient to induce the C1q tagging of synapses, it may not be absolutely necessary for this outcome, as synaptic C1q buildup is also stimulated by amyloidosis and can occur in early AD brain tissue prior to the detection of AT8-immunoreactive phospho-Tau (see Figure 5D). (2) Although conspicuously localized at synapses in tauopathy mice, C1q is also deposited at non-synaptic sites on neurons and possibly other cells suggesting that other structures could be targets of microglial engulfment or complement-mediated damage.

How does C1q recognize Tau-damaged synapses, and is there a link between changes in the PSD composition and postsynaptic C1q binding? During development, complement-dependent synaptic pruning by microglia is modulated by synaptic activity such that active synapses are preserved and less active synapses are removed (Schafer et al., 2012), implying that complement can "detect" morphofunctional changes at the synapse. In principle, Tau-driven changes in the PSD could lead to a signal for recruitment of C1q. In this context, it will be critical to identify the synaptic receptor(s) for C1q. The co-immunoprecipitation of C1q with PSD-95 indicates that C1q associates with the PSD-95-scaffolded complex, where it might bind to a transmembrane protein of as yet unknown identity. Alternatively, pathogenic Tau might trigger the exposure of lipid molecules like phosphatidylserine, a well-characterized "eat me" signal and C1q ligand on apoptotic cells (Païdassi et al., 2008), perhaps through local non-lethal activation of caspase-3 in spines (Ertürk et al., 2014; Györffy et al., 2018).

We observed increased synaptic material engulfed by microglia in Tau-P301S mice, and this engulfment could be reduced with a C1q-blocking antibody. A key question is whether microglia engulf intact synapses or simply phagocytose synaptic debris that has been shed from injured neurons. Our neuron-microglia co-cultures suggest that, at least *in vitro*, microglia are able to remove excitatory synapses without apparent damage to the dendrite or loss of inhibitory synapses. Recently, using light sheet fluorescence microscopy and CLEM, microglia have been shown to "nibble" and phagocytose presynaptic, but not postsynaptic, terminals during development (Weinhard et al., 2018). In this context, it is interesting that we detect C1q in normal brain mostly at presynaptic locations, whereas the buildup of C1q in Tau-P301S brain takes place mainly at postsynaptic sites, where it may promote the removal of Tau-damaged postsynapses and spines. The rescue of synapse density following anti-C1q treatment of Tau-P301S mice is more consistent with a role of microglia in removing viable synapses rather than simply clearing debris.

It is remarkable that a single injection of C1q-blocking antibodies not only reduced microglial engulfment of synapses *in vivo* but also enhanced synapse density 5 days post injection. This rescue of synapse number suggests an ongoing dynamic turnover of synapses in the Tau-P301S brain that can be rebalanced in a favorable direction by neutralization of C1q. Despite amelioration of synapse density, a single injection of anti-C1q did not improve LTP. As revealed in our proteomic analysis, however, there are multiple other changes induced by tauopathy in the PSD that might not be fixed by neutralization of C1q. More optimistically, perhaps longer treatment with C1q antibodies is needed to obtain benefit beyond the structural saving of synapses. The above notwithstanding, complement activation and synapse loss seems to be a feature of multiple neurological diseases, including progranulin-deficient frontotemporal dementia (Lui et al., 2016), virus-induced memory impairment (Vasek et al., 2016), glaucoma (Stevens et al., 2007), and CNS lupus (Bialas et al., 2017); therefore, a therapeutic approach that blocks C1q and microglial eating of synapses is worth exploring for a range of neurodegenerative disorders.

STAR★METHODS

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SUPPLEMENTAL INFORMATION

Supplemental Information includes eight figures and three tables and can be found with this article online at https://doi.org/10.1016/j.neuron.2018.10.014.

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AUTHOR CONTRIBUTIONS

B.D. designed the study, performed experiments, and analyzed results. M.A.H. and B.A.F. performed the proteomic and genomic informatics analyses. A.D.M. and J.K. planned and performed all EM experiments. W.J.M., Z.J., T.W., K.S., H.N., and B.C. characterized Tau-P301S mice and helped with experiments. O.F. analyzed the AD patient tissue histology. V.G. and R.A.D.C. performed brain MRI experiments. C.B. performed and analyzed the MS experiments. J.E.H. performed electrophysiology and helped with experimental design. M.S. directed the study. B.D. and M.S. wrote the manuscript. All authors read and edited the manuscript.

DECLARATION OF INTERESTS

All authors were full-time employees of Genentech during the course of these studies, except for A.D.M. and J.K.

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STAR★METHODS

KEY RESOURCES TABLE

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Antibodies		
ApoJ/Clu	Santa Cruz	RRID: AB_673567
beta-actin (HRP-conjugate)	Sigma	RRID: AB_262011
beta-tubulin (HRP-conjugate)	Cell Signaling Technology	RRID: AB_1950376
C1q	Abcam	RRID: AB_10711046
C1q	Abcam	RRID: AB_2732849
C1q	Abcam	RRID: AB_1280770
C1q	Dako	RRID: AB_2335698
C1q-blocking antibody	This paper	N/A
CD11b-APC	BD Pharmingen	RRID: AB_10897015
CD11b/c Alexa405-conjugate	Novus	RRID: AB_714950
CD68	Bio-Rad	RRID: AB_322219
Ezrin	Santa Cruz	RRID: AB_783303
GABAA receptor alpha1	Synaptic Systems	RRID: AB_2619929
Gephyrin	Synaptic Systems	RRID: AB_887719
Gephyrin	Synaptic Systems	RRID: AB_887717
GFAP	Dako	RRID: AB_10013382
GFP	Abcam	RRID: AB_300798
Homer1	Cell Signaling	RRID: AB_10858221
Homer1	Synaptic Systems	RRID: AB_2120990
Iba1	Synaptic Systems	RRID: AB_2493179
Iqgap1	Santa Cruz	RRID: AB_2249072
Iqgap1	Abcam	RRID: AB_11157518
Iqgap2	Santa Cruz	RRID: AB_2264816
Iqsec2	Thermo Fisher Scientific	RRID: AB_2634921
Kalirin	Millipore	RRID: AB_310380
MAP2	Abcam	RRID: AB_2138153
NeuN	Millipore	RRID: AB_2298772
Prrt1	NeuroMab	RRID: AB_2531894
PSD-95	Cell Signaling Technology	RRID: AB_10828082
PSD-95	Millipore	RRID: AB_2092365
PSD-95	NeuroMab	RRID: AB_2307331
Shank3	NeuroMab	RRID: AB_2315920
Synapsin	Synaptic Systems	RRID: AB_2622240
Synaptophysin	Synaptic Systems	RRID: AB_1210382
SynGAP1	Thermo Fisher Scientific	RRID: AB_2287112
Tau	Synaptic Systems	RRID: AB_993042
Tau AT100	Thermo Fisher Scientific	RRID: AB_223652
Tau AT180	Thermo Fisher Scientific	RRID: AB_223649
Tau AT8	Thermo Fisher Scientific	RRID: AB_223647
Tau HT7	Thermo Fisher Scientific	RRID: AB_2314654
Tau p396	GeneTex	RRID: AB_10620290
Tau pS202	Cell Signaling Technology	11834
Tau pS400/T403/S404	Cell Signaling Technology	11837

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REAGENT or RESOURCE	SOURCE	IDENTIFIER
Tau pS404	Abcam	RRID: AB_10561457
Tau pT181	Cell Signaling Technology	12885
Tau pT205	GeneTex	RRID: AB_10721986
Tau-13	BioLegend	RRID: AB_2565341
Tau5	Abcam	RRID: AB_1603723
Trio	Bethyl Labs	RRID: AB_2620465
Biological Samples		
Human brain tissue	Folio Biosciences	N/A
Chemicals, Peptides, and Recombinant Proteins		
Phallacidin	Abcam	ab143532
Phalloidin iFluor 488 reagent	Abcam	ab176753
Purified human C1q	Complement Tech	A099
Normal goat serum	Cell Signaling	5425
Paraformaldehyde	Electron Microscopy Sciences	15710
ProLong Gold Antifade Mountant	Invitrogen	P36930
Lipofectamine 2000	Thermo Fisher Scientific	11668019
NbActiv4 medium	BrainBits	Nb4-500
Experimental Models: Cell Lines		
HIP NSC	MTI-GlobalStem	GSC-4311
Experimental Models: Organisms/Strains		
Mouse: Prnp.Tau.P301S.tg.B6N	The Jackson Laboratory	Stock No: 02481
Mouse: Thy-1.PrP.hu.APP.hu.PS2.tg.B6	Roche	Richards et al., 2003
Rat: Sprague Dawley	Charles River	Strain code: 400
Oligonucleotides		
ON-TARGETplus non-targeting pool siRNA	Dharmacon	D-001810-10-05
ON-TARGETplus SMARTpool siRNA Iqgap1	Dharmacon	L-082950-02-0005
ON-TARGETplus SMARTpool siRNA Iqgap2	Dharmacon	L-115738-00-0005
ON-TARGETplus SMARTpool siRNA Iqsec2	Dharmacon	L-113951-00-0005
ON-TARGETplus SMARTpool siRNA Iqsec3	Dharmacon	L-100631-02-0005
ON-TARGETplus SMARTpool siRNA Kalrn	Dharmacon	L-099410-02-0005
ON-TARGETplus SMARTpool siRNA Trio	Dharmacon	L-087912-02-0005
Accell siRNA Trio	Dharmacon	E-087912-00-0010
Accell siRNA Kalrn	Dharmacon	E-099410-00-0010
Accell siRNA Iqgap2	Dharmacon	E-115738-00-0010
Accell siRNA Iqgap1	Dharmacon	E-082950-00-0010
Accell siRNA Iqsec2	Dharmacon	E-113951-01-0010
Accell red non-targeting siRNA	Dharmacon	D-001960-01-20
Recombinant DNA		
Plasmid: Tau-P301S-IRES-GFP	This paper	N/A
Plasmid: GFP	This paper	N/A
Software and Algorithms		
Prism 7	GraphPad	https://www.graphpad.com/scientific-software/prism/
ImageJ	NIH	https://imagej.nih.gov/ij/
Imaris 8.3.1	Bitplane	http://www.bitplane.com/imaris
InCell 6000 analysis software	GE Life Sciences	https://www.gelifesciences.com/en/de/shop/cell-imaging-and-analysis/high-content-analysis-systems/software/in-carta-image-analysis-software-p-00329

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REAGENT or RESOURCE	SOURCE	IDENTIFIER
MATLAB	MathWorks	https://www.mathworks.com/products/matlab.html
Huygens Deconvolution	Scientific Volume Imaging	https://svi.nl/HuygensDeconvolution
TopScan	CleverSys	http://cleversysinc.com/products/software/topscan/
Image Lab Software	Bio-Rad	http://www.bio-rad.com/en-mx/product/image-lab-software
Image Studio Software	LI-COR	https://www.licor.com/bio/products/software/image_studio/index.html
Other		
Nanoparticles	Precision Nanosystems	https://www.precisionnanosystems.com/our-technology/spark

CONTACT FOR REAGENT AND RESOURCE SHARING

Further information and requests for resources and reagents should be directed to and will be fulfilled by the Lead Contact, Morgan Sheng (sheng.morgan@gene.com)

EXPERIMENTAL MODEL AND SUBJECT DETAILS

Mice

In all experiments, animals were randomly assigned to experimental or treatment groups and matched for age and littermate controls. All mice were female. All behavioral scorings, electrophysiological, microscopic and histological analyses were carried out by experimenters blind to genotype and treatment groups. Tau-P301S mice were originally purchased from Jackson Lab (B6N.Cg-Tg(Prnp-MAPT**P301S)PS19Vle/J, stock #02481). Tau-P301S mice express mutant human Tau driven by prion promoter. PS2/APP mice that co-express human APP with the Swedish mutation K670N/M671L and human presenilin 2 with the N141I mutation, driven by Thy1 and PrP promoters were described previously (Richards et al., 2003). Non-transgenic littermates were used as controls. Human Tau alleles were bred to heterozygosity, PS2/APP mice to homozygosity in all mice used in this study. All animal studies were authorized and approved by the Genentech Institutional Animal Care and Use Committee.

Human subjects

Post mortem brain tissue (superior frontal gyrus) was obtained from patients with putative AD as indicated by cognitive evaluation scores (“EAD”) or pathology-confirmed AD (“AD”) along with age-matched controls from Folio Biosciences. Human samples were procured with Ethics Committee approval and written informed consent. Patient data for all samples used in this study is provided in Table S3.

Primary rat neurons and microglia

Rat neuronal cultures were prepared from hippocampi of rat embryos on embryonic day 18 (E18) or E19 and plated on poly-D-lysine and laminin coated coverslips at a density of 90,000/24-well dish and cultured in NbActiv4 medium (BrainBits). 50% of the medium was exchanged with fresh medium weekly. Proteins were knocked down using Accell siRNA (Dharmacon) or siRNAs (Dharmacon) that were packed into nanoparticles (Precision Nanosystems) and added to the cultures. All siRNAs were pooled and contained 4 unique siRNAs per gene. Neurons were usually treated 5-7 days with the siRNA before analysis. Neurons were transfected with Lipofectamine 2000 (Invitrogen) according to manufacturer’s manual. For primary microglial cultures, embryonic E18-E19 or postnatal (P1-P2) pups were decapitated and forebrain was triturated with a 5 ml serological pipette and the homogenate was spun at 300xg for 5 min. The supernatant was discarded and the pellet was resuspended with a 1 ml pipette and filtered through a 75 μm filter. Two brains were cultured per 175 cm^2 flask in 40 ml DMEM + 10% fetal bovine serum (FBS). After 24 h flasks were rinsed with Hank’s Balanced Salt Solution (HBSS, Thermo Fisher Scientific) and new media was added. Cultures were grown for an additional 7-9 days before microglia were shaken off the astrocyte feeder layer on a rocking platform for 1 h, pelleted and resuspended in NbActiv4 medium and added to neurons in a 1:3 (microglia:neuron) ratio. Primary cells from male and female pups were used.

Human iPSC-derived neurons

Human iPSC derived neural stem cells (NSC), male, were purchased from MTI-Global Stem. NSC was maintained and differentiated following the manufacturer’s instructions. NSC maintenance media: DMEM/F12/Neurobasal, 1X GS22, 20 ng/ml bFGF, 20 ng/ml EGF, 20 ng/ml BDNF, 0.11 mM beta-mercaptoethanol, 0.5 mM Glutamax. NSC differentiation media: DMEM/F12/Neurobasal, 1X GS21, 1X N2, 5μg/ml Cholesterol, 1 mM Creatine, 100 μM ascorbic acid, 0.5 mM cAMP, 20 ng/ml GDNF, 20 ng/ml BDNF, 1 μg/ml Laminin, 0.5 mM Glutamax. Human NSCs were plated to PDL and Laminin-coated 96 well plates at 40,000/cm^2 in differentiation media. 50% of media was changed biweekly for the duration of the experiment. After culturing for 6-8 weeks, the neurons were used for experiments.

METHOD DETAILS

Brain immunohistochemical and immunofluorescence staining

Mice were deeply anesthetized and transcardially perfused with phosphate-buffered saline (PBS), followed by 4% paraformaldehyde (PFA) in PBS for fixation. Brains were post-fixed in 4% PFA overnight, then cryoprotected in 30% sucrose. Alternatively, the right hemi-brain was drop fixed for 48 h at 4°C in 4% PFA for immunohistochemistry and the left forebrain was frozen and later homogenized. Brains were either embedded in a solid matrix and sectioned at 30 μm by NeuroScience Associates or were sectioned in house at 35μm using a sliding microtome. Sections were preserved in cryoprotectant (30% glycerol, 30% ethylene glycol, and PBS) and stored at −20°C.

For fluorescent immunostainings, free-floating sections were rinsed in PBS, followed by PBS with 0.2% Triton X-100 (PBST), blocked with blocking buffer (5% goat serum, 5% bovine serum albumin (BSA) in PBST) and incubated overnight with primary antibodies in blocking buffer. For PSD-95 immunostaining, sections were pre-incubated with 1% Triton X-100 to make the PSD accessible before staining. Secondary antibodies in blocking buffer were incubated for 2-3h at room temperature, extensively washed in PBST and PBS and mounted using ProLong Diamond medium (ThermoFisher).

For 3’-Diaminobenzidine (DAB) immunostaining, free-floating sections were rinsed in Tris-buffered saline (TBS), treated with hydrogen peroxide in TBST (TBS plus Triton X-100), and blocked with TBST plus goat serum before overnight incubation at room temperature with primary antibody diluted in TBST. Binding was detected with biotin-conjugated secondary antibodies followed by avidin-biotin-HRP complex (Vectastain elite ABC kit, Vector Laboratories) and developed with diaminobenzidine tetrahydrochloride. CD68-immunostained sections were counterstained with a Nissl stain (0.05% thionine/0.08 M acetate buffer, pH 4.5).

Tissue imaging and analysis

For total immunofluorescence analysis, brains sections were scanned on a Hamamatsu NanoZoomer S60. Mean fluorescence intensity of region of interest (ROI) was measured in MATLAB (MathWorks). Histological samples were imaged using the Leica SCN400 system equipped with automated loader and software (Leica Microsystems). Quantification of DAB staining and enlarged dark cluster areas was performed using morphometric-based methods in MATLAB, as previously described (Le Pichon et al., 2013). Color thresholds in RGB and HSV space were used to identify the positive staining. The percentage DAB-positivity for hippocampal ROIs were calculated by normalizing the positive pixel area to section area.

All sections were averaged from three to seven sections/animal. All images, segmentation overlays, and data were reviewed by a pathologist.

For microglia engulfment analysis, brain sections were imaged on the Zeiss LSM 780 taking up to 35 z-stacks at 0.39 μm steps with a 100x oil objective. CD68-positive lysosomes were 3D-reconstructed using the surface rendering function in Imaris 8.3 and Synapsin or PSD-95 puncta inside the CD68+ lysosomes were quantified. 6-9 microglia within the hippocampal CA1 region were analyzed per mouse.

For STED super-resolution imaging, brain sections were stained for Synapsin, PSD-95 and C1q and hippocampal CA1 regions were imaged on a Leica SP8 STED 3x microscope using a 100x objective at a pixel size of 15 nm/pixel. Images were deconvoluted using the Huygens package and distanced between the cluster centers were measured in ImageJ (NIH). For C1q-Synapsin co-localization, STED images were analyzed using the ImageJ plugin ComDet 0.3.6.1 and automatically detected clusters were considered as co-localized if distance between spot centers was ≤ 200 nm.

Immunoelectron microscopy (immunoEM) and analysis

Fixed brains were cut in 1 mm thick sagittal slices. Whole brain slices as well as isolated, ~1 mm^3 blocks of hippocampus tissue were then processed for immunoEM according to the Tokuyasu method (Slot and Geuze, 2007). Briefly, tissue slices and blocks were rinsed in PBS and PBS with 0.15% glycin, embedded in 12% gelatin in 0.1 M PB and cryoprotected with 2.3 M sucrose in 0.1 M PB. Hippocampal tissue blocks were mounted in random orientation on aluminum pins and frozen in liquid nitrogen. From the brain slices, the hippocampus was excised and cut in an anterior and posterior half, each ~1 mm^3 in size. Each block was then mounted on an aluminum pin such that sagittal hippocampus sections could be cut with known ventral-dorsal orientation, and frozen in liquid nitrogen.

From these blocks, semithin cryosections, 300-400 nm thick, were cut at -100 °C, placed on glass coverslips, labeled with primary antibodies, fluorescent secondary antibodies and DAPI, and examined in a DeltaVision wide-field fluorescence microscope. Ultrathin cryosections for immunoEM were cut at -120 °C, placed on copper carrier grids and sequentially incubated on PBS at 37 °C to dissolve gelatin, then at room temperature on primary and secondary antibody solutions, followed by 10 nm Protein A-gold particles (Cell Microscopy Core, University Medical Center Utrecht, the Netherlands). Incubation times were 1 h at room temperature for Tau-13, AT8 and secondary antibodies, and overnight at 4 °C for C1q clone 4.8 antibodies. Blocking solutions contained either BSA, or a mixture of fish skin gelatin and acetylated BSA (BSA-c, from Aurion). Final staining of the sections was performed with uranyl acetate (UAc) followed by a UAc-methylcellulose mixture. For correlative light-electron microscopy (CLEM) (van Rijnsoever et al., 2008), sections on grids were additionally labeled with a fluorescent secondary antibody and DAPI, imaged in a DeltaVision fluorescence microscope, and subsequently rinsed and contrasted with UAc and UAc-methylcellulose for electron microscopy.

For the quantitative analysis of the distribution of Tau-P301S in the synapse, the CA1 region was retraced using CLEM images of Tau-13 immunogold-labeled ultrathin sagittal cryosections of 3 mice per genotype. All gold particles labeling Tau-P301S in axon terminus and spine of 20-24 synapses per Tau-P301S mouse were counted and their distance to the synaptic membranes was measured. The distances of all Tau-13 gold particles were pooled, ranked in 30 nm-classes, and the frequencies, divided by the total number of synapses, were plotted.

For the quantitative analysis of the presence of C1q at the synapse, the molecular layer from the dentate gyrus region was retraced using CLEM images of C1q immunogold labeled, ultrathin sagittal cryosections of 3 mice per genotype. Per mouse, 69-108 synapses were systematically screened and per synapse the presence of C1q immunogold particles was scored as either on presynaptic or postsynaptic or on both pre- and postsynaptic compartments.

FACS-sorting of microglia

Microglia were isolated from nonTg and Tau-P301S mouse forebrains and quality controlled as described previously (Srinivasan et al., 2016). All steps were carried out at 4°C. Briefly, after perfusion with ice cold PBS, cortex and hippocampi were dissected and triturated to single cell suspension. After obtaining the cell pellet after Percoll gradient step, cells were resuspended in 1 ml of Hibernate A-LF and incubated with Cd11b-APC (BD Pharmingen, clone M1/70, #562690) at 1:200. Cells were then washed and resuspended in 4 ml of Hibernate together with 2 μl of DAPI (1mg/ml). Animals were always processed in pairs (1 nonTg and 1 Tau-Tg) and sorted simultaneously in two BD FACS ARIA sorters. Myeloid cells ($DAPI^-$:$CD11b^+$) were collected, pelleted at 300xg, 5 min and lysated in 100 μl lysis buffer (50 mM Tris, 150 mM NaCl, 1% TritonX-100, 0.2% SDS) supplemented with protease inhibitors (cOmplete mini, Roche). Cell debris was pelleted (10,000g, 5 min) and equal volumes were used for Western blotting.

Brain slice electrophysiology

400 μm thick slices of hippocampus were prepared with a vibrating sectioning system (Leica, Germany). Slices were recorded while bathed in oxygenated artificial cerebrospinal fluid (ACSF) containing (in mM): 127 NaCl, 2.5 KCl, 1.3 $MgSO_4$, 2.5 $CaCl_2$, 1.25 Na_2HPO_4, 25 $NaHCO_3$, and 25 glucose. During slice preparation ice-cold ACSF was used with the MgSO4 concentration elevated to 7 mM, NaCl replaced with 110 mM choline, and with the addition of 11.6 mM Na-ascorbate and 3.1 mM Na-pyruvate. Standard field recordings were performed as previously described (Hanson et al., 2013). Schaffer collateral inputs were electrically stimulated and EPSPs and population spikes were recorded from stratum radiatum and stratum pyramidale of CA1, respectively. LTP was induced using 1, 2, or 3 bouts of burst stimulation (separated by 20 s), each burst consisting of 5 pulses at 100 Hz.

Stereotaxic antibody injection

The blocking anti-C1q antibody was produced using the variable domain sequences of the previously described anti-C1q M1 antibody (Hong et al., 2016a) on a mouse IgG2a backbone with the effector-attenuating substitutions L234A, L235A, and P329G (Lo et al., 2017). 9-months-old female mice were anesthetized with 3% isoflurane and placed on a stereotaxic apparatus (Stoelting) for surgery. All injections were performed with a 10 μl syringe (Hamilton Company) with a pulled glass pipette tip glued to the end of needle, and a syringe pump (Stoelting Quintessential stereotaxic injector). The injection coordinate was 1.25 mm anterior, −1.82 mm lateral, 1.75 mm ventral to bregma. 2 μl of anti-C1q or isotype control antibody (both 28 mg/ml) were injected at a rate of 0.5 μl/min. The needle was removed 2 minutes after completion of the injection and mice were put back in their cage for 5 days before analysis.

Brain MRI

Longitudinal MRI study was performed on Tau-P301S and nonTg female mice (n=10 each group) by imaging animals at 6, 9 and 12 months of age using a 9.4T Varian/Agilent (Agilent Technologies) system with a 30 mm inner-diameter transmit/receive volume coil. The animals were anesthetized using 3% isoflurane and placed in a custom-built head holder with a tooth bar. During imaging, anesthesia was maintained at 1.5% isoflurane. Rectal temperature was monitored and maintained at 37 ± 1°C using a feedback system with warm air (SA Instruments). T2-weighted images were acquired using a multi-spin echo sequence with TR=6000 ms, TE ranged from 10 – 80 ms for 8 echoes with a spacing of 10 ms, 56 axial slices of 0.3 mm, FOV 19.2 mm x 19.2 mm, matrix size 128 x 96 (zero-filled to 128 x 128), 2 signal averages, with a scan time of 19 min.

Behavioral tests

Locomotor activity

The experimental room was illuminated, providing ~300 lux in the middle of the locomotor chambers. PAS open-field recording systems (San Diego Instruments) consisted of transparent thermoplastic cages (width, 40.5 cm; length, 40.5 cm; height, 38 cm) surrounded by a locomotor frame and a rearing frame fitted with infrared beams to record horizontal locomotor activity (3 cm above the floor) and rearing activity (7.5 cm above the floor), respectively. The total number of beam breaks for both locomotor activity and rearing was calculated for a total of 15 min.

Morris water maze

Mice were trained in the Morris water maze as previously described (Hanson et al., 2014). Briefly, mice were trained to locate a hidden platform (15 cm) submerged 1.5 cm below the water for five consecutive days. Mice received two training sessions per day, each session consisted of three trials (10 min between trials), and each session was separated by 3-4 hours, for a total of 10 hidden sessions. The platform location remained the same throughout the hidden-platform training, but the drop location was changed semi-randomly between trials. Mice had 60 sec to locate the platform during training. Mice that did not find the platform were guided to it and placed on it for 10 sec. A spatial probe trial was performed at the beginning of day 4 (session 7, probe 1) 24 h (probe 2) and 72 h (probe 3) after the completion of hidden-platform training. The platform was removed, and mice were allowed to swim in the pool for 60 sec. For cued visible-platform training, the platform (15 cm diameter) was submerged (1.5 cm) but visibly cued with a striped mast (15 cm in height). Over 2 consecutive days, mice received one or two training sessions per day separated by 3–4 h, for a total of 3 cued sessions. Each session consisted of two training trials with an intertrial interval of 10 min. For each session, the platform was moved to a new location, and the drop location was changed semi-randomly between trials. Escape latencies, swim paths, swim speeds, percentage of time spent in each quadrant, and platform crossings (in a zone 200% the size of the platform) were recorded using a ceiling-mounted camera and tracked using CleverSys TopScan software.

Expression analysis of complement genes in control and AD brain RNA

Data were analyzed as described in "Gene Set Scores" Methods of Friedman et al. (2018). Briefly, "Complement gene set score" for each sample (indicated equivalently by color strip below main plot and by y-axis of Figure S5B) was calculated as the average, taken over all 51 genes, of the mean-centered log2-expression for each sample. The heatmaps in Figure S5A show Z-score-normalized gene expression, which was performed separately within each dataset. Samples within control and AD groups for each dataset were sorted by first principal component of the Z-score matrix, and genes were ordered by their correlation with the overall gene set score vector across both datasets.

Neuronal immunofluorescence staining and image analysis

Immunofluorescence was performed as previously described (Dejanovic et al., 2014) with modification. Briefly, cultured cells were fixed with 4% PFA in PBS for 12 min and blocked in blocking buffer (5% goat serum, 5% BSA, 0.2% Triton X-100 in PBS). Primary antibodies were incubated either for 2 h at room temperature or overnight at 4°C. After three washing steps, Alexa-conjugated (-405, -488, -567 or -647) secondary antibodies (ThermoFisher) were incubated for 1-2 h before mounting the cover slips.

For spine or synaptic cluster analysis, stained cells were imaged in a confocal Zeiss LSM 780 microscope with a 60x oil objective taking five z-stacks. Maximum intensity projections were created in ImageJ (NIH). Spine density was quantified using NeuronStudio 0.9.92 (Rodriguez et al., 2008) or through manual counting in ImageJ. Synaptic density along dendrites was measured in ImageJ and normalized to the length on the dendritic segment. The experimenter was blinded to the treatment of analyzed neurons.

To measure C1q labeling of synapses in culture, transfected neurons and microglia were co-cultured for three days prior to the analysis. Alternatively, human native C1q (1μg/ml; Complement Technology) was directly added to the culture medium of transfected neurons and cultures were further incubated for 1h. Cells were washed twice with Neurobasal medium and once with PBS before fixation. Immunostained cultures were imaged on a Zeiss LSM 780 with a 100x oil objective taking three-four z-stacks (0.4μm step size). A macro was written and used for the analysis. Maximum intensity projections were created and the neuronal GFP signal was thresholded and used as a mask to detect PSD-95 puncta in the transfected neurons. To analyze the percentage of C1q+ PSD-95 puncta, binary images were created and C1q puncta were considered as synaptically localized if at least one pixel colocalized with a PSD-95 cluster. All C1q puncta in a field and all PSD-95 puncta within the transfected neuron were examined.

C1q binding to human neurons

Human native C1q (Complement Technology), pre-treated with control IgG or C1q-blocking antibody, was incubated with neuronal cultures for 1 h. After washing three times, neurons were fixed, immunostained and imaged on InCell 6000 (GE Life Sciences) using 20X lens with 9 fields per well, 6 wells per treatment condition. Image analysis was done using InCell 6000 analysis software. C1q images were thresholded to subtract background and exclude large aggregate artifacts and total intensity of the C1q immunostaining was analyzed.

Preparation of postsynaptic density

PSD fractions were isolated as described (Peng et al., 2004). Briefly, brain regions of interest were dissected and homogenized in homogenization buffer (5 mM HEPES (pH 7.4), 1mM $MgCl_2$, 0.5mM $CaCl_2$) supplemented with phosphatase (PhosStop, Roche) and protease inhibitors (cOmplete mini, Roche) using a Teflon homogenizer. After 1,400 g, 10 min centrifugation the supernatant was pelleted at 13,800 g, 10 min and pellet was re-suspended in 0.32M Tris-buffered sucrose and ultra-centrifuged into 1.2, 1.0, 0.85M sucrose gradient at 82,500 g, 2h. Synaptosome fraction was solubilized with 1% Triton X-100 and pellet was collected at 32,800 g for 20 min. The pellet was extracted in 0.5% Triton X-100 and re-centrifuged at 200,000 g for 1h to save the final PSD pellet.

Immunoprecipitation

Mouse hippocampi were homogenized in lysis buffer (5mM HEPES (pH 7.4), 1 mM $MgCl_2$, 0.5 mM $CaCl_2$), supplemented with phosphatase (PhosStop, Roche) and protease inhibitors (cOmplete mini, Roche). Postnuclear fraction was centrifuged at 10,000 g for 10 min. The pellet was solubilized in lysis buffer + 0.5% Triton X-100, cleared at 10,000 g for 10 min, and incubated with PSD-95 or Gephyrin antibodies overnight. Immuncomplexes were collected using magnetic Protein-G beads (Dynabeads, ThermoFisher) and rotated for 2 h. After three washes, beads were boiled in SDS loading buffer and analyzed by Western blotting.

Western blotting

Protein samples were boiled in reducing SDS loading buffer and ran on NuPAGE Bis-Tris gradient gels (Invitrogen). Gels were transferred to nitrocellulose membranes (Bio-Rad) and blocked with 5% milk powder or BSA. Primary antibodies were incubated overnight at 4°C and secondary HRP-conjugated (abcam) or IRDye fluorescent (Li-Cor, Rockland) antibodies for 1-2 h at room temperature. Immunoreactivity was detected on ChemiDoc XRS+ (Bio-Rad) or Odyssey infrared Imaging System (Li-Cor) and analyzed using Image Lab Software (Bio-Rad) or Image Studio Software (Li-Cor).

Mass spectrometry analysis of PSD

Six PSD samples (n = 3 Tau-P301S, n = 3 non-Tg littermate controls) were separated via SDS-PAGE and divided into 7 gel bands. Gel bands were separately digested using trypsin as previously described (Kirkpatrick et al., 2013), lyophilized, and reconstituted in 0.1% formic acid with 2% acetonitrile. Samples were separated using C18 online chromatography as described before introduction into an LTQ Orbitrap Elite mass spectrometer (Thermo Scientific). Full MS data were acquired in FT for 375–1600 m/z with a 60 000 resolution. The 15 most abundant ions found in the full MS were selected for MS/MS through a 2-Da isolation window.

Mass spectral data were assigned to peptides using a concatenated target-decoy database consisting of mouse sequences from UniProt (version 2015_04) using Mascot (Matrix Science, Boston, MA, USA) using a 50 ppm precursor ion mass tolerance, 0.8 Da fragment ion tolerance, a fixed proprionamide modification on all cysteines and a variable oxidation modification on methionine, fully-tryptic specificity and a maximum of two missed cleavages. Peptide-spectral matches were first filtered to a false discovery rate (FDR) of 5% at the peptide level using a linear discriminant approach. Filtered peptides were subsequently filtered to a 2% protein FDR.

Quantification of peak areas corresponding to filtered peptides using the VistaGrande XQuant algorithm (Bakalarski et al., 2008; Kirkpatrick et al., 2013). Quantification was performed on a band-by-band basis considering any peptide identifications in that band and its neighboring bands only. Results were further filtered to a confidence score of 83 or greater.

Linear mixed-effects modeling of peptide abundances was carried out as described previously (Kirkpatrick et al., 2013) combining all samples from each individual into a single condition while separating individuals. Peak areas were normalized on a sample-by-sample basis to ensure equal median peak intensities across all samples.

Changes in peptide abundance were calculated and tested for statistical significance. Median fold-changes and p-values from the pairwise Tau-P301S vs. non-Tg control comparisons (shown in Table S1) were used for downstream analysis. Proteins were considered to be differentially abundant if they had a log2 fold-change > 0.5, and an adjusted p-value < 0.05.

Gene Ontology analysis

For Gene Ontology analysis, all peptides were mapped to Entrez gene IDs, and the "universe" of genes used for analysis was restricted to only the proteins that were detected by MS and available for differential abundance testing. Two gene sets were used for Gene Ontology analyses: genes whose proteins were significantly more abundant in the Tau-P301S PSD samples, and those which were significantly more abundant in the non-Tg littermate control PSD samples (i.e. significantly less abundant in the Tau-P301S). Conditional hypergeometric tests for ontologies of Molecular Function, Biological Processes, and Cellular Compartment were performed using the Bioconductor R package GOstats, version 2.44.0. Further stringent filters were applied to the results, requiring p-values < 0.01, and at least 4 differentially abundant genes overlapping a given ontology category. The top 8 results ranked by odds-ratios for each direction, is represented in Figure 2C for the Biological Processes Ontology.

QUANTIFICATION AND STATISTICAL ANALYSIS

Quantification is described in Method Details and in figure legends. Number of samples is shown in the figure and described in the figure legends. Unless otherwise stated, statistical analysis was performed using GraphPad Prism 7 and the used statistical tests are defined in the figure legend. In all instances, statistical significance was defined as follows: n.s. - not significant, *$p < 0.05$, **$p < 0.01$, ***$p < 0.001$. Data throughout the paper is displayed as mean ± SEM. No outlier analysis was performed or data points excluded. Sample size was chosen according to that used for similar experiments in previously published literature. Experimenters were blinded whenever possible to experimental condition during data acquisition or quantification.